

Lie Groups and Lie Algebras: Curvatures and Representations

AUBERT Voalaza Mahavily Romuald (aubert@aims.ac.za)
International Centre for Theoretical Physics (ICTP)

Supervised by: Professor Claudio Arezzo and Shubham Sinha
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Abstract

Lie groups form a natural framework for describing continuous symmetries in geometry, analysis, and mathematical physics. Their associated Lie algebras capture the infinitesimal structure of the group, and the exponential map links these two perspectives. This thesis investigates the interplay between the algebraic and geometric aspects of Lie groups, with a particular emphasis on left-invariant Riemannian metrics, Cartan connections, and representation theory.

After reviewing the necessary background on smooth manifolds, Riemannian geometry, and Lie theory, we study the Lie-theoretic and Riemannian exponential maps, comparing their properties and highlighting cases where they coincide or differ. Using Milnor's framework, we perform explicit curvature computations for Lie groups with left-invariant metrics. We also explore the representation theory of complex semisimple Lie algebras, presenting examples for $\mathfrak{sl}(2, \mathbb{C})$, $\mathfrak{sl}(3, \mathbb{C})$, and more general cases.

The main goal is to study representations of Lie algebras, which can contain more information about Lie groups and are often easier to analyze than the groups themselves. In this thesis, we focus only on complex cases, but we expect to be able to address the real ones later using realification or complexification theory.

This project will allow me to learn about Lie groups, Lie algebras, their curvature, and their representations.

Keywords: Lie theory, representation theory, Lie groups, Lie algebras, differential geometry.

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Introduction

Lie groups and Lie algebras play a major role in modern mathematics, making them one of the most standard languages for describing continuous symmetries in geometry, analysis, and mathematical physics. They stem from the exploratory research of Sophus Lie, who in the late 19-th century discovered that they provide an appropriate mathematical context for describing both global and infinitesimal symmetries. A Lie group is both a differentiable manifold and a group, with the property that the two structures are compatible such that both the multiplication and inversion operations are smooth maps. Instead of being defined only by its group operation, a Lie group can alternatively be described by its associated ‘Lie algebra,’ which captures the infinitesimal behaviour near the identity element. This picture of interplay between these two perspectives, is captured by the exponential map.

Beyond their intrinsic beauty, Lie groups and their representations play a decisive role in diverse areas of mathematics and physics. In differential geometry, they arise naturally as transformation groups of manifolds, with applications to the study of homogeneous spaces and symmetric spaces [Helgason \(2024\)](#). The theory of semisimple Lie algebras, in particular, provides a systematic way to study high-dimensional symmetry via the representation theory of finite-dimensional modules.

From a geometric perspective, left-invariant Riemannian metrics on Lie groups yield rich families of spaces whose curvature can often be computed explicitly, revealing deep structural properties [Milnor \(1976\)](#). This connection has been developed extensively in the context of symmetric spaces and Cartan geometries, where Lie groups act transitively by isometries, and their isotropy subgroups encode the local geometry. The advantage of having a bi-invariant metric is that geodesics can be understood using the Lie exponential map, which is not always true when only a left-invariant metric is present. Then the Cartan connection can change the role of the Levi-Civita connection so it still holds.

The motivation for this thesis is to understand Lie groups and their geometry. We aim to study Lie groups via their Lie algebras, and also through the representations of these Lie algebras, that can help as to understand some class of Lie groups.

The structure of the thesis is as follows. Chapter [1](#) reviews the necessary background on smooth and Riemannian manifolds. Chapter [2](#) introduces Lie groups, Lie algebras, and their representations. Chapter [3](#) focuses on the exponential map, its consequences, and its interaction with Riemannian geometry, including a discussion of Lie groups with left-invariant metrics and Cartan connections. Chapter [4](#) develops the representation theory of semisimple Lie algebras, with examples from $\mathfrak{sl}(2, \mathbb{C})$, $\mathfrak{sl}(3, \mathbb{C})$ and more general complex semisimple Lie algebras. This goal applies to all Lie groups and Lie algebras, both real and complex, but here we start with the complex semisimple ones. Finally, Chapter [5](#), we end with some remarks.

1. On Smooth Manifolds and Riemannian Manifolds

In this section, we start with important facts about smooth manifolds that will be useful for the next part of this dissertation. It will be mostly about smooth manifolds, smooth maps submanifolds and Riemannian manifolds. The main references of this section are [Lee \(2003\)](#) and [Do Carmo and Flaherty Francis \(1992\)](#).

1.1 Smooth manifolds and smooth maps

1.1.1 Definition (Smooth manifold). A smooth manifold M of dimension n , consists of

- (i) a topological space M , Hausdorff and second countable;
- (ii) a smooth atlas which is collection of pairs $(U_i, \phi_i)_{i \in I}$, $\phi_i : U_i \rightarrow \tilde{U}_i$; such that:
 - ϕ_i is a homeomorphism where $(\tilde{U}_i)_{i \in I}$ are open subsets of $\mathbb{R}^n$ (namely they are charts) and $(U_i)_{i \in I}$ is an open cover of M ;
 - for any $i, j \in I$: $\phi_j \circ \phi_i^{-1} : \phi_i[U_i \cap U_j] \rightarrow \phi_j[U_i \cap U_j]$ are smooth $\mathcal{C}^\infty$.
- (iii) the atlas is maximal, in the sense that if (U, ϕ) and for any $i \in I$, $\phi \circ \phi_i^{-1}, \phi_i \circ \phi^{-1}$ are smooth on $\phi_i[U_i \cap U], \phi[U \cap U_i]$, then (U, ϕ) must belong to the atlas.

For such above notation, in the next, we describe a manifold $M = (M, (U_i, \phi_i, \tilde{U}_i)_{i \in I}, n)$ of dimension n . So, in the next of this dissertation, M is always with these structure, unless we mention the opposite. In practice to check that a Hausdorff, second countable topological space is a manifold, it is enough to see that it is locally homeomorphic to $\mathbb{R}^n$ and the composition of the maps are smooth. Indeed, we can neglect the maximality of atlas because we can always construct an unique maximal one from a given atlas.

 **Notice.** One can define similarly the complex manifold like smooth manifold above, space locally homeomorphic to $\mathbb{C}^n$ and the transition maps required to be holomorphic. One can see directly is any complex manifold of dimension m is smooth manifold of dimension $n = 2m$ and if a smooth manifold of dimension n is also complex manifold, n must be even.

1.1.2 Lemma. For any smooth atlas $(U_\alpha, \phi_\alpha)_{\alpha \in A}$,

$$\mathcal{A} = \{(U, \phi) \text{ chart such that for any } \alpha \in A : \phi_\alpha \circ \phi^{-1}, \phi \circ \phi_\alpha^{-1} \text{ are smooth}\} = (U_i, \phi_i)_{i \in I}$$

is the unique maximal atlas that contains $(U_\alpha, \phi_\alpha)_{\alpha \in A}$.

Proof. Clearly, $A \subset I$. To prove that it is an atlas, the only thing that we need to check is the case where both of i, j is in $I \setminus A$. We want to show that $\phi_j \circ \phi_i^{-1}$ is locally smooth on $\phi_i[U_i \cap U_j]$, then smooth. Let $\tilde{p} \in \phi_i[U_i \cap U_j]$ and $p = \phi_i^{-1}(\tilde{p})$. There is $\alpha \in A$ that contains p . Then $p \in \phi_i[U_i \cap U_j \cap U_\alpha]$ and $\phi_j \circ \phi_i^{-1} = (\phi_j \circ \phi_\alpha^{-1}) \circ (\phi_\alpha \circ \phi_i^{-1})$ is smooth on $\phi_i[U_i \cap U_j \cap U_\alpha]$. It is maximal by definition as $A \subset I$. The uniqueness is by similar reasoning. Assume, $(U_j, \phi_j)_{j \in J}$ and $(U_k, \phi_k)_{k \in K}$ such that $A \subset J \cap K$. By symmetry, we will prove only $(U_k, \phi_k)_{k \in K} \subset (U_j, \phi_j)_{j \in J}$. Fix an arbitrary $k \in K$ and let's show that $\phi_k \circ \phi_j^{-1}$ and $\phi_j \circ \phi_k^{-1}$, so $(U_k, \phi_k) \in (U_j, \phi_j)_{j \in J}$. We pick again $\tilde{p} \in \phi_k^{-1}(U_k \cap U_j) \in U_\alpha$, for some $\alpha \in A$ to get that $\phi_j \circ \phi_k^{-1}$ is smooth near to $\tilde{p}$. $\square$

1.1.3 Remark. Every open subset U of a smooth manifold is trivially itself smooth manifold because we can just restrict each chart on its domain intersects U .

1.1.4 Definition (Smooth map). Let M and N be two manifolds whose the smooth atlas are $(U_i, \phi_i)_{i \in I}$ and $(V_j, \varphi_j)_{j \in J}$. The map $F : M \rightarrow N$ is said to be smooth if

for each $i \in I, j \in J$, the map $\varphi_j \circ F \circ \phi_i^{-1}$ is smooth on $\phi_i(U_i \cap F^{-1}[V_j])$.

Again, as we mostly work on charts which can cover the manifolds that is atlas simpler to work with but this definition requires as to check for all possible charts (maximal atlas). Therefore, this version is sometimes less practical. There is the next equivalent definition is important.

1.1.5 Definition (Smooth map). With the assumptions in Definition 1.1.4, F is smooth if

for each $p \in M$, there are (U, ϕ) , chart in M , $p \in U$ and (V, φ) , chart in N , $F(p) \in V$ such that $\varphi \circ F \circ \phi^{-1}$ is smooth on $\phi(U \cap F^{-1}[V])$.

Clearly Definition 1.1.4 implies 1.1.5. Conversely if we have 1.1.5, let's try to prove 1.1.4. Let $i \in I, j \in J$. To show $\varphi_j \circ F \circ \phi_i^{-1}$ is smooth on $\phi_i(U_i \cap F^{-1}[V_j])$, recall, from calculus that being C^∞ for map between subset of $\mathbb{R}^n$ is a local property. Then, it is enough to prove that for each $\tilde{p} \in \phi_i(U_i \cap F^{-1}[V_j])$, $\varphi_j \circ F \circ \phi_i^{-1}$ is C^∞ near to $\tilde{p}$. Consider $p = \phi_i^{-1}(\tilde{p})$ and apply 1.1.5 to get charts $(U, \phi), (V, \varphi)$ containing p and $F(p)$ such that $\varphi \circ F \circ \phi^{-1}$ is smooth on $\phi(U \cap F^{-1}[V])$. From the structure of the smooth manifolds, we have $\phi \circ \phi_i^{-1}$ and $\varphi_j \circ \varphi^{-1}$ are respectively smooth on $\phi_i[U \cap U_i]$ and $\varphi[V_j \cap V]$. Thus $\varphi_j \circ F \circ \phi_i^{-1} = (\varphi_j \circ \varphi^{-1}) \circ (\varphi \circ F \circ \phi^{-1}) \circ (\phi \circ \phi_i^{-1})$ is smooth on $\phi_i[U \cap U_i \cap F^{-1}[V_j]]$, which is an open neighbourhood of $\tilde{p}$.

It is clear that the smoothness is a local property.

1.1.6 Remark. Smoothness implies continuity. It is because this is known for the case subsets of $\mathbb{R}^n$ and the fact the continuity is local property. Indeed, if $p \in M$, we choose the charts (U_i, ϕ_i) and (V_j, φ_j) that contain p and $F(p)$ with $\varphi_j \circ F \circ \phi_i^{-1}$ is smooth on $\phi_i(U_i \cap F^{-1}[V_j])$. Then $\varphi_j \circ F \circ \phi_i^{-1}$ is continuous on $\phi_i(U_i \cap F^{-1}[V_j])$ and $F = \varphi_j \circ (\varphi_j \circ F \circ \phi_i^{-1}) \circ \phi_i$ is continuous on $U_i \cap F^{-1}[V_j]$, which is open neighbourhood of p .

1.1.7 Remark (On product). It is straightforward to see that for any smooth manifolds M and N , $M \times N$ is also smooth manifolds. The charts are just the products. Now, we have the following results.

- (i) If we have another smooth manifolds M', N' with $f_1 : M \rightarrow M'$ and $f_2 : N \rightarrow N'$ are both smooth. The product map $f_1 \times f_2 : M \times N \rightarrow M' \times N'$, $(m, n) \mapsto (f_1(m), f_2(n))$ is also smooth. If $(p, q) \in M \times N$, there are $(U_i, \phi_i), (V_j, \varphi_j), (W_k, \rho_k)$ and (O_l, δ_l) respectively charts in M, M', N and N' , that contain respectively $p, f_1(p), q$ and $f_2(p)$ such that $\varphi_j \circ f_1 \circ \phi_i^{-1}$ and $\delta_l \circ f_2 \circ \rho_k^{-1}$ are smooth. Hence $(\varphi_j \times \delta_l) \circ (f_1 \times f_2) \circ (\phi_i^{-1} \times \rho_k^{-1}) = (\varphi_j \circ f_1 \circ \phi_i^{-1}) \times (\delta_l \circ f_2 \circ \rho_k^{-1})$ is smooth.
- (ii) We have also a diffeomorphism¹ $M \times N \rightarrow N \times M, (x, y) \mapsto (y, x)$.
- (iii) The projections are smooth. Again, the continuity is known in the level of topology. For $(p, q) \in M \times N$, there are (U_i, ϕ_i) and (V_j, φ_j) charts in M and N containing p and q . $\phi_i \circ \pi_M \circ (\phi_i \times \varphi_j)^{-1}$ and $\varphi_j \circ \pi_N \circ (\phi_i \times \varphi_j)^{-1}$ are just projections which are trivially smooth.
- (iv) For any smooth maps $f : X \rightarrow M$ and $g : X \rightarrow N$, $f \times g : X \rightarrow M \times N, x \mapsto (f(x), g(x))$ is also smooth. Hence the universal property.
- (v) Fixing $p \in M$, the inclusion map $i_p : N \rightarrow M \times N$ defined by $i_p(y) = (p, y)$ is smooth. Let

¹smooth homeomorphism, whose inverse also smooth.

$y \in N$ and consider the charts (U_i, ϕ_i) and (V_j, φ_j) in M and in N , which contains p and y . Then $(U_i \times V_j, \phi_i \times \varphi_j)$ is a chart that contains $i_p(x)$ and $(\phi_i \times \varphi_j) \circ i_p \circ \varphi_j^{-1}$ is the inclusion $z \mapsto (\phi_i(p), z)$ which is trivially smooth. The same applies if we fix $q \in N$ and $j_q(x) = (x, q)$.

As in $\mathbb{R}^n$, we know that the differentiability is a local property and the restriction of differentiable map is also differentiable. These properties still work in general smooth manifold.

1.1.8 Proposition. With the assumptions in Definition 1.1.4, we have.

- (a) If F is smooth then for any open subset U of M , the restriction $F|_U : U \rightarrow N$ is smooth.
- (b) If any point of M admits open neighbourhood U which $F|_U$ is smooth, then, F is smooth

Proof. (a) is straightforward by the second definition, as restriction of differentiable map is differentiable in $\mathbb{R}^n$. For (b), pick $p \in M$; by assumption $p \in U$ and $F|_U : U \rightarrow N$ is smooth. Then there is $k \in I, j \in J$ such that $p \in U_k$, $F(p) \in V_j$ and $\varphi_j \circ F|_U \circ (\phi_k|_U)^{-1} = \varphi_j \circ F \circ (\phi_k|_U)^{-1}$ is smooth. So, it is enough to prove that $(U \cap U_k, \phi_k|_U) \in (U_i, \phi_i)_{i \in I}$. To see it, we just need to use the fact $\phi_i[U_k \cap U_i \cap U]$ is open subset of $\phi_i[U_i \cap U]$ and again restriction of differentiable map is differentiable in $\mathbb{R}^n$. The same for $\phi_i[U_k \cap U_i \cap U] \subset \phi_i[U_i \cap U]$, so, we get $\phi_i \circ (\phi_k|_U)^{-1}$ and $(\phi_k|_U) \circ \phi_i^{-1}$. $\square$

We have defined objects (the collections of smooth manifolds) and morphisms (the smooth map), we expect to get a category. So we want to check if the composition is well-defined.

1.1.9 Remark. The good news is the composition of smooth maps is smooth. It is straightforward. Indeed, consider $M \xrightarrow{F} N \xrightarrow{G} P$ be two smooth maps (denote $(W_k, \delta_k)_{k \in K}$ atlas on P). Let $i \in I$ and $k \in K$. To prove $\delta_k \circ G \circ F \circ \phi_i^{-1}$ is smooth on $\phi_i[U_i \cap F^{-1} \circ G^{-1}[W_k]]$, let's do it locally. Pick $\tilde{p} \in \phi_i[U_i \cap F^{-1} \circ G^{-1}[W_k]]$ so $p = \phi_i(\tilde{p}) \in U_i \cap F^{-1} \circ G^{-1}[W_k]$ and $F(p) \in G^{-1}[W_k]$. Then there exists, (V_j, φ_j) chart in N that contains $F(p)$ such that $\delta_k \circ G \circ \varphi_j^{-1}$ and $\varphi_j \circ F \circ \phi_i^{-1}$ are smooth. Then $\delta_k \circ G \circ F \circ \phi_i^{-1} = (\delta_k \circ G \circ \varphi_j^{-1}) \circ (\varphi_j \circ F \circ \phi_i^{-1})$ is smooth on $\phi_i[U_i \cap F^{-1} \circ G^{-1}[W_k] \cap F^{-1}[V_j]] \ni \tilde{p}$. Then, smooth manifolds form a category. The isomorphisms are the diffeomorphisms.

1.2 Derivations, Vector fields, Tangent spaces, Differential maps

Let A be an algebra over a field $\mathbb{K}$. A derivation D on A is its linear endomorphism satisfying the product rules, that is $D(fg) = fD(g) + D(f)g$. The algebra with which we shall mainly deal is the algebra of smooth functions on a smooth manifold M , denoted by $\mathcal{C}^\infty(M)$.

1.2.1 Definition. For a smooth manifold (M, n) , we mean by a vector field on M is just a derivation on $\mathcal{C}^\infty(M)$. A derivation at p is just $X_p : \mathcal{C}^\infty(M) \rightarrow \mathbb{R}, f \mapsto X(f)(p)$ for some X vector field on M .

We will denote $\mathfrak{X}(M)$ the set of vector fields on M . The set $T_p M := \{X_p : X \text{ vector field on } M\}$ is a vector space of dimension n .

1.2.2 Definition. For a smooth map $F : M \rightarrow N$, the differential of F at a point $p \in M$ is a linear map

$$\begin{aligned} (dF)_a : T_p M &\longrightarrow T_{F(p)} N \\ X_p &\longmapsto (dF)_a(X_p) = Y_{F(p)}, \end{aligned}$$

where Y is vector field such that $Y_{F(p)}(g) = X_p(g \circ F)$ ² for any $g \in \mathcal{C}^\infty(N)$.

²Such vector field Y exists.

Another view of tangent space T_p is set of vector v where there is smooth curve α on M , defined around p , where $\alpha'(0) = v$ and $\alpha(0) = p$. In this point of view,

$$(dF)_p(\alpha'(0)) = \left. \frac{d}{dt} \right|_{t=0} F \circ \alpha(t).$$

1.3 Submanifolds: embedded vs immersed

At the topological level, any subset X of a topological space M inherits a topology that makes the inclusion a topological embedding. This topology is natural in M .

Now, when a smooth structure is added, we require more conditions to be able to get a smooth manifold structure on X , so that it is nice, in the sense that we will try to define a submanifold.

Intuitively, we expect that any chart (U_i, ϕ_i) of M should induce a chart on X , namely that $(X_i = U_i \cap X, \phi_i|_{X_i})$ should produce a smooth atlas for X . But, in the end, it is not exactly that we need; rather, we will reformulate it in the sense that we can put a smooth structure on X that is induced (in a natural way) from the structure on M . However, not every subset of M satisfies such a condition.

1.3.1 Definition (Embedded Submanifold). Let $(M, (U_i, \phi_i, \tilde{U}_i)_{i \in I}, n)$ is manifold. $X \subset M$ is said a submanifold of M , of dimension $k \in \mathbb{N}$, $k \leq n$ if X can be covered by $(U_i)_{i \in A}$ ($A \subset I$) such that

$$\forall i \in A : \quad \phi_i(X \cap U_i) = \mathbb{R}_k \cap \tilde{U}_i, \quad (1.3.1)$$

where $\mathbb{R}_k = \{x \in \mathbb{R}^n : x_j = 0 \text{ for } j > k\}$ homeomorphic to $\mathbb{R}^k$.

1.3.2 Definition (Smooth immersion). A smooth map $M \xrightarrow{F} N$ is a smooth immersion if for each $p \in M$, $(dF)_p$ is injective. Moreover, if F is topological embedding, we say F is smooth embedding. Dually, F is smooth submersion if $(dF)_p$ is surjective for each $p \in M$.

1.3.3 Lemma. Under the assumption in Definition 1.3.1 and (1.3.1), let $\mathbb{R}_k \xrightarrow{\varepsilon_k} \mathbb{R}^k$ be the natural homeomorphism. Let $X_i = U_i \cap X$ and $\rho_i = \varepsilon_k \circ \phi_i|_{X_i}$. We have $(X_i, \rho_i)_{i \in A}$ is smooth atlas on X and the inclusion is a smooth immersion.

From now on, if we think X as submanifold of M , with the topology induced, X is also smooth manifold with the atlas constructed in Lemma 1.3.3.

1.3.4 Proposition. Let M and X be two smooth manifolds such that $X \subset M$ (the topology on X is the induced one from M). If the inclusion map is a smooth immersion then X is submanifold of M . and the smooth structure on X is the same than we constructed in Lemma 1.3.3.

1.3.5 Example. Let $M \xrightarrow{F} N$ be an embedding. We expect $F(M)$ to be identifiable to M . Clearly, they can be diffeomorphic as they are homeomorphic and we can push forward the charts in M via F . In this case, the inclusion $F(M) \hookrightarrow N$ becomes a composition of diffeomorphism and embedding then embedding. Hence $F(M)$ is an (embedded) submanifold of N .

Now, how about if we have only smooth immersion, can we still have $F(M)$ is submanifold? First to be able (easy) to identify $F(M)$ with M , via F , we require F to be injective. Even though issues arise at the topological level, such cases are still interesting, which is called immersed submanifold. Namely, we can change the topology in $F(M)$ (not necessarily the induced from N), with specific smooth structure to get the inclusion $F(M) \hookrightarrow N$ becomes a smooth immersion. The topology and smooth structure are both the push forward from M via F that make F diffeomorphic onto $F(M)$.

1.3.6 Definition (Immersed Submanifold). Let M and X be two smooth manifolds such that $X \subset M$ (the topology on X is not necessarily the one induced from M). If the inclusion map is a smooth immersion, we say that X is an immersed submanifold of M .

Some example will be seen in Lie subgroups section, for Example 2.1.8.

The following propositions are the combination of some theorems or propositions in Lee (2003).

1.3.7 Proposition. Let $M \xrightarrow{F} N$ be a smooth map.

- (a) For $p \in M$, if $(dF)_p$ is injective (resp surjective) then F is a smooth immersion (resp submersion) near to p . Also, if $(dF)_p$ is isomorphism, F is diffeomorphism near to p .
- (b) If F is bijective and local diffeomorphism then it is diffeomorphism. Also smooth immersion and smooth submersion implies local diffeomorphism.

1.3.8 Proposition. Let $(M, (U_i, \phi_i, \tilde{U}_i)_{i \in I}, m) \xrightarrow{F} (N, (V_i, \varphi_i, \tilde{V}_i)_{i \in I}, n)$ be smooth map with constant rank r , that is

$$\forall p, q \in M : \quad \text{rank}((dF)_p) = \text{rank}((dF)_q) = r.$$

Then the followings hold.

- (a) For each $p \in M$ there is $i \in I$ and $j \in J$ such that $p \in U_i, F(p) \in V_j$ and

$$\hat{F}(x_1, \dots, x_m) := \varphi_j \circ F \circ \phi_i((x_1, \dots, x_m) = (x_1, \dots, x_r, 0, \dots, 0).$$

- (b) F is smooth immersion if it is injective. Also it smooth submersion if it is surjective.

1.3.9 Remark. Item (a) is a necessary and sufficient condition for a smooth map to have constant rank. More generally, it is enough that $\hat{F}$ is linear.

1.4 Quotient Manifolds

This part is almost from Lee (2003). The goal of this section is to note that not every quotient topology naturally gives a manifold structure and to state some important facts in which case this quotient also has a smooth structure. First, a continuous map $f : X \rightarrow Y$ is called proper if the preimage of a compact subset of Y is compact in X . For a group action, if G acts on a manifold M , we denote M/G the set of orbits with the quotient topology, called the orbit space. Assume now that G is a topological group. The action is proper if it is continuous and the map $G \times M \rightarrow M \times M, (g, p) \mapsto (g \cdot p, p)$ is a proper map.

1.4.1 Theorem (Quotient Manifold Theorem). *Suppose G is a Lie group acting smoothly, freely, and properly on a smooth manifold M . Then the orbit space M/G is a topological manifold of dimension equal to $\dim M - \dim G$, and has a unique smooth structure with the property that the quotient map $\pi : M \rightarrow M/G$ is a smooth submersion.*

The proof can be found in Lee (2003), together with many interesting discussions about it.

We are interested in these results in order to see, in Section 2.1, how the quotient of a Lie group can be described. Lee discusses the Homogeneous Space Construction Theorem, which provides a useful fact, but only in the case of Lie groups.

1.5 Riemannian manifolds

This section is mostly taken from [Do Carmo and Flaherty Francis \(1992\)](#). We now collect some of the important facts that we will need.

1.5.1 Definition (Riemannian manifold). A *Riemannian manifold* is a pair (M, g) , where M is a smooth manifold and g assigns to each point $p \in M$ smoothly an inner product

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

on the tangent space $T_p M$, varying smoothly with $p \in M$.

1.5.2 Definition (Affine connection). An *affine connection* ∇ on a smooth manifold M is a map

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M), \quad (X, Y) \mapsto \nabla_X Y$$

satisfying for all $X, Y, Z \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$:

- $\nabla_{fX+Y} Z = f\nabla_X Z + \nabla_Y Z$ (linearity in the first argument),
- $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$ (linearity in the second argument),
- $\nabla_X (fY) = X(f)Y + f\nabla_X Y$ (Leibniz rule).

A vector field along a curve $\gamma : I \rightarrow M$ is a smooth map $X : I \rightarrow TM$ such that, for any $t \in I$, $X(t) \in T_{\gamma(t)} M$.

1.5.3 Proposition. Let M be a smooth manifold with an affine connection ∇ . There exists a unique correspondence which associates to a vector field V along the smooth curve $c : I \rightarrow M$ another vector field $\frac{DV}{dt}$ along c , called the covariant derivative of V along c , such that:

- (a) $\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt}$.
- (b) $\frac{D}{dt}(fV) = \frac{df}{dt}V + f\frac{DV}{dt}$, where W is a vector field along c and f is a smooth function on I .
- (c) If V is induced by a vector field $Y \in \mathfrak{X}(M)$, i.e., $V(t) = Y(c(t))$, then $\frac{DV}{dt} = \nabla_{\dot{c}(t)} Y$.

A vector field along curve $\gamma : I \rightarrow M$ is called *parallel* when $\frac{DV}{dt} = 0$, for all $t \in I$.

1.5.4 Theorem (Existence and uniqueness of the Levi-Civita connection). Let (M, g) be a Riemannian manifold. There exists a unique affine connection ∇ on M such that

- ∇ is *metric-compatible*, i.e. for all vector fields X, Y, Z ,

$$X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z),$$

- ∇ is *torsion-free*, i.e. for all X, Y ,

$$\nabla_X Y - \nabla_Y X = [X, Y].$$

This connection ∇ is called the *Levi-Civita connection*.

A parametrized curve $\gamma : I \rightarrow M$ is a geodesic if $\frac{D}{dt}(\frac{d\gamma}{dt}) = 0$ at t_0 , for any $t_0 \in I$.

The important fact is that, after solving the ODE in local coordinates, for each $p \in M$ and $v \in T_p M$, there exists a unique geodesic $\gamma_v(t)$, defined in a neighborhood of 0, such that $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$. For small v , this γ_v can be defined at $[0, 1]$. Thus, we can define the exponential map on a small open set U containing p ,

$$\begin{aligned} \text{Exp}_p : T_p M \supseteq U &\longrightarrow M \\ v &\longmapsto \text{Exp}_p(v) = \gamma_v(1). \end{aligned}$$

A Riemannian manifold is called geodesically complete if the maximal domain of definition of all geodesics is $\mathbb{R}$. One can show that compact manifolds are complete. Moreover,

1.5.5 Theorem (Hopf–Rinow Theorem [Alexandrino et al. \(2015\)](#)). Let (M, g) be a connected Riemannian manifold and $p \in M$. The following statements are equivalent:

- (i) $\exp_p$ is globally defined, that is, $\exp_p : T_p M \rightarrow M$;
- (ii) M is geodesically complete;
- (iii) (M, dist) is a complete metric space;

$$\text{dist}(p, q) := \inf \left\{ \int_a^b \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt : \gamma(a) = p, \gamma(b) = q, \gamma \text{ piecewise smooth} \right\}.$$

- (iv) Every closed bounded set in M is compact.

If M satisfies any (hence all) of the above items, each two points of M can be joined by a minimal geodesic segment. In particular, for each $x \in M$ the exponential map $\exp_x : T_x M \rightarrow M$ is surjective.

Next, we recall the concept of curvature.

1.5.6 Definition (Riemann curvature tensor). Let (M, g) be a Riemannian manifold with connection ∇ . The *Riemann curvature tensor* is the $(1,3)$ -tensor defined by:

$$\forall X, Y, Z \in \mathfrak{X}(M) : R(X, Y)Z = \nabla_Y \nabla_X Z - \nabla_X \nabla_Y Z + \nabla_{[X, Y]} Z.$$

1.5.7 Definition (Sectional curvature). Let (M, g) be a Riemannian manifold, and let $\sigma \subset T_p M$ be a 2-dimensional subspace spanned by linearly independent vectors $u, v \in T_p M$. The *sectional curvature* of σ at p is defined by:

$$\kappa(\sigma) = \frac{g(R(u, v)v, u)}{\|u\|^2 \|v\|^2 - g(u, v)^2}.$$

1.5.8 Definition (Ricci curvature). The *Ricci curvature* is the $(0,2)$ -tensor defined by taking the trace of the Riemann curvature tensor:

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y).$$

1.5.9 Definition (Scalar curvature). The *scalar curvature* is the function S on M defined as the trace of the Ricci tensor:

$$\rho = \text{tr}_g(\text{Ric}) = \sum_{i=1}^n \text{Ric}(e_i, e_i), \quad \text{where } \{e_i\} \text{ is a local orthonormal basis.}$$

1.5.10 Theorem. [Bonnet–Myers] Let (M^n, g) be a complete Riemannian manifold of dimension n . Suppose there exists a constant $\delta > 0$ such that the Ricci curvature satisfies

$$\operatorname{Ric}(v, v) \geq (n - 1) \delta \|v\|^2 \quad \text{for all } v \in TM.$$

Then:

- (i) M is compact,
- (ii) the diameter of M satisfies $\operatorname{diam}(M) \leq \frac{\pi}{\sqrt{\delta}}$,
- (iii) and the fundamental group $\pi_1(M)$ is finite.

2. Lie groups and Lie Algebras

In this chapter, we discuss general aspects of Lie groups and Lie algebras, with some examples. One important point that we address is how Lie algebras arise from Lie groups.

2.1 Lie groups

2.1.1 Definition (Lie group). A Lie group G consists of

- (i) a group $G = (G, m, inv, e_G)$, where m is the binary operation and inv the inverse mapping;
- (ii) a smooth structure on G ;
so that:
- (iii) G is a smooth manifold;
- (iv) m and inv are smooth maps.

Thus, a Lie group is simply a group that is also smooth manifold where the group operations are smooth.

 **Notice.** This definition is about the smooth manifolds (real) but one can think also the complex Lie groups where we require to be complex manifold and the operations to be holomorphic. From now on, unless otherwise specified, by a Lie group we mean a real Lie group. ¹

After $\mathbb{R}^n$, the one most important example is the linear group $\mathbf{GL}(n, \mathbb{R})$ and some of its subgroups.

2.1.2 Example (Linear Lie group). Identifying $\mathbf{M}_n(\mathbb{R})$ to $\mathbb{R}^{n^2}$, the determinant map, being polynomial, is continuous. Then $\mathbf{GL}(n, \mathbb{R}) = \mathbf{M}_n(\mathbb{R}) \setminus \det^{-1}(\{0\})$ is open subset of $\mathbf{M}_n(\mathbb{R})$. So it is manifold of dimension n^2 . By Remark 1.1.3, it is manifold. Also, it is group under the multiplication of polynomial in the entries; the inverse mapping is given by rational functions with nonzero determinant in the denominator, hence both operations are smooth. Hence $\mathbf{GL}(n, \mathbb{R})$ is a Lie group.

We will prove later that any closed subgroup of $\mathbf{GL}(n, \mathbb{R})$ is also Lie group.

2.1.3 Example (Product). As the product of smooth manifolds is smooth manifold and also group, then we should expect that the product of Lie group is again Lie groups. It is. Indeed, the smoothness of the inverse and the binary operation is the direct consequence of Remark part (i) and (ii).

2.1.4 Definition (Lie group morphism). A Lie group morphism is just a map between Lie group that is smooth and group homomorphism.

 **Notice.** For this definition, continuity is enough, we will prove it later at Proposition 3.1.7 but for some of the examples, we assume it as fact.

The composition of Lie group morphisms is clearly Lie group morphism. Then we can talk about the isomorphism of Lie group and category of Lie groups $\mathcal{LG}$.

2.1.5 Example (Projections). We want to see that the product of Lie groups verifies the universal property with projections. By definition, the projections are Lie group morphisms². It is also clear the unique map that for any pair of Lie group morphisms $g : M \rightarrow G$ and $h : M \rightarrow H$, the unique map that commutes with the projections $g \times h$ is also Lie group morphism. Hence it is the product in $\mathcal{LG}$.

¹That is fair because any complex manifold is smooth manifold.

²From Remark 1.1.7 part (iii).

2.1.6 Proposition. Any Lie group morphism $G \xrightarrow{F} H$ has constant rank. Consequently, any bijective Lie group morphism is an isomorphism.

Proof. It is straightforward to see that the translations (left and right) are diffeomorphisms and

$$F(L_p(x)) = L_{F(p)}(F(x)) \implies (dF)_p \circ (dL_p)_{e_G} = (dL_{F(p)})_{e_H} \circ (dF)_e.$$

Then $(dF)_p = (dL_{F(p)})_{e_H} \circ (dF)_e \circ (dL_p)_{e_G}^{-1}$ has same rank than $(dF)_e$ because composition with an isomorphism don't change rank. The last is a consequence of Propositions 1.3.7 and 1.3.8. $\square$

2.1.7 Definition (Lie Subgroup). A subgroup H of Lie group G is said immersed Lie subgroup if it is an immersed submanifold of G and the binary operation is smooth map from $H \times H$ to H . Moreover, if the topology that makes H submanifold of G is the one induced from G (embedded submanifold), we say that it is embedded Lie subgroup.

Notice. After building the theory, one can prove is that if H is subgroup of G and H is an immersed submanifold of G , then, it has to be immersed Lie subgroup. The proof is by following the sketch in Spivak (1970), Problem 20 of chapter 10.

For us, Lie subgroup is always immersed one, unless mention the opposite. We will explain later why.

2.1.8 Example. Let $\mathbb{R} \xrightarrow{F} S^1 \times S^1, F(x) = (e^{2\pi i x}, e^{\pi\sqrt{2}ix})$. It is injective and smooth immersion as $F'(p) \neq 0$ for any $p \in \mathbb{R}$. Then $F(\mathbb{R})$ is an immersed submanifold of $S^1 \times S^1$. We know that S^1 is multiplicative group(Lie group) and clearly F is (Lie) group homomorphism. Then $F(\mathbb{R})$ is a Lie immersed subgroup of $S^1 \times S^1$.

Moreover, noticing that for any $\varepsilon > 0$, there exists a positive integer n such that the decimal part of $\frac{n\sqrt{2}}{2}$ is less than ε , and that $|e^{ix} - 1| \leq |x|$, we get $|F(n) - F(0)| \leq 2\pi \left(\frac{n\sqrt{2}}{2} - \left\lfloor \frac{n\sqrt{2}}{2} \right\rfloor \right) < 2\pi\varepsilon$. Then there is a sequence $(y_n)_{n \geq 0} \subset F(\mathbb{N}_{\geq 1})$ that converges to $F(0)$. Since there is no sequence in $\mathbb{N}_{\geq 1}$ that converges to 0, the map $F(\mathbb{R}) \xrightarrow{F^{-1}} \mathbb{R}$ is not continuous, i.e., F is not an embedding. Thus $F(\mathbb{R})$ is not an embedded submanifold.

2.1.9 Proposition. Let H be open subgroup of a Lie group G . Then it is embedded Lie subgroup of G and union of connected components of G .

Proof. By Remark 1.1.3, H has a chart induced from G . Then, in local coordinates, the inclusion $\hat{i}$ is just the inclusion map, so its derivative is injective at every $h \in H$. Hence, H is an embedded Lie subgroup. Noticing that $G = \bigcup_{g \in G} L_g(H)$, $H = G \setminus \bigcup_{g \in G \setminus H} L_g(H)$, and that each L_g is a homeomorphism, it follows that H is also closed. Since H is both closed and open, it is a union of connected components. Indeed, if H intersects a connected component C , then $H \cap C$ is a nonempty, closed, and open subset of the connected space C . Therefore, $H \cap C = C$, and hence $C \subset H$. $\square$

It is not hard to see that an open neighbourhood V of e_G generates an open subgroup of G . Indeed, $\tilde{V} = V \cup (-V)$ is open, and $\tilde{V} + {}^3\tilde{V} = \bigcup_{v \in \tilde{V}} L_v(\tilde{V})$ is open. Similarly, $V_k = \underbrace{\tilde{V} + \dots + \tilde{V}}_{k \text{ times}}$ is open for every k . Then $\langle V \rangle$ is open. Using a similar argument and the fact that $e_G \in V, -V, \tilde{V}, L_v(\tilde{V})$, and V_k , we also obtain that if V is connected, then $\langle V \rangle$ is connected.

2.1.10 Proposition. Let G connected Lie group and U is an open and $e_G \in U$. Then U generates G .

Proof. From above discussion with Proposition 2.1.9. $\square$

³The symbol $+$ denotes the group operation.

Also, one can show that the identity component is an open normal subgroup of G , and that it is diffeomorphic to every connected component.

2.1.11 Proposition. Let H be an immersed Lie subgroup of G . Then H is closed if and only if H is embedded.

Proof. Assume that H is embedded. We show that H is closed in G . First, we claim that H is open in $\overline{H}$. Indeed, let $p \in H$. Since H is an embedded submanifold, there exists a chart (U, ϕ) of G with $p \in U$ such that $\phi(U \cap H) = \tilde{U} \cap \mathbb{R}^k$. Taking closures in U and in $\tilde{U}$, we obtain $\phi(U \cap \overline{H}) = \tilde{U} \cap \mathbb{R}^k = \phi(U \cap H)$, hence $U \cap \overline{H} = U \cap H$. Thus, $U \cap \overline{H}$ is an open neighbourhood of p in $\overline{H}$, and therefore H is open in $\overline{H}$. Next, for each $x \in \overline{H}$, the left translation L_x restricts to a homeomorphism of $\overline{H}$, so the sets $xH = L_x(H)$ are open and dense in $\overline{H}$. Now let $a \in \overline{H}$. Since $a \in aH$ and aH is dense in $\overline{H}$ while H is open in $\overline{H}$, we have $H \cap aH \neq \emptyset$. Hence $aH = H$, which implies $a \in H$. Therefore $\overline{H} \subseteq H$, so H is closed.

The converse is found in [Lee \(2003\)](#). □

2.1.12 Theorem. Any closed subgroup H of a Lie group G is always Lie subgroup.

The proof can be found in [Spivak \(1970\)](#), theorem 15 of chapter 10. By Proposition [2.1.11](#), it is embedded.

This theorem allows us to understand what the linear Lie groups are. They are the subgroups of $\mathbf{GL}(n, \mathbb{R})$ ⁴ which are Lie groups with the topology inherited from $\mathbf{GL}(n, \mathbb{R})$ (and also from $\mathcal{M}_n(\mathbb{R})$). That is, they are the embedded Lie subgroups of $\mathbf{GL}(n, \mathbb{R})$. Hence, the linear Lie groups are the closed subgroups of $\mathbf{GL}(n, \mathbb{R})$.

As examples, there are

$$\begin{aligned} \mathbf{SL}(n, \mathbb{R}) &= \{A \in \mathbf{GL}(n, \mathbb{R}) : \det(A) = 1\}, \\ \mathbf{O}(n, \mathbb{R}) &= \{A \in \mathbf{GL}(n, \mathbb{R}) : A^T A = I_n\}, \\ \mathbf{SO}(n, \mathbb{R}) &= \mathbf{O}(n, \mathbb{R}) \cap \mathbf{SL}(n, \mathbb{R}), \\ \mathbf{O}(p, q) &= \{A \in \mathbf{GL}(p+q, \mathbb{R}) : A^T I_{p,q} A = I_{p,q}\}, \\ \mathbf{SO}(p, q) &= \mathbf{O}(p, q) \cap \mathbf{SL}(p+q, \mathbb{R}), \\ \mathbf{Sp}_{2n} \mathbb{R} &= \{A \in \mathbf{GL}(2n, \mathbb{R}) : A^T J_n A = J_n\}, \end{aligned}$$

where $I_{p,q} = \text{diag}(1, \dots, 1, -1, \dots, -1)$ (p positive and q negative) and $J_n = (a_{ij})_{1 \leq i,j \leq 2n}$ with $a_{n+i,i} = 1, a_{i,n+1} = -1$ for $i = 1, \dots, n$ and zero otherwise. We may add some examples later. There are also some complex Lie groups defined in a similar way. Just a remark: the unitary groups $\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are defined like $\mathbf{O}(n, \mathbb{R})$ and $\mathbf{SO}(n, \mathbb{R})$ but over $\mathbb{C}$, replacing the transpose by the conjugate transpose. However, they are not complex Lie groups, but real Lie groups.

2.1.13 Lemma. Consider the smooth map,

$$\begin{aligned} m_k : \prod_{i=1}^{i=k} G &\longrightarrow G \\ (g_1, \dots, g_k) &\longmapsto m_k((g_1, \dots, g_k)) = m(g_1, m(g_2, \dots, m(g_{k-1}, g_k) \dots)) = g_1 \cdot g_2 \cdots g_k. \end{aligned}$$

Let us agree that whenever we write $\alpha'(0) \in T_p G$, we mean that α is a curve in G such that $\alpha(0) = p$.

⁴We can assume all subgroups of $\mathbf{GL}(n, \mathbb{R})$ because $\mathbf{GL}(n, \mathbb{C})$ can be viewed inside $\mathbf{GL}(2n, \mathbb{R})$.

In this case,

$$(dm_k)_{(p_1, \dots, p_k)}(\alpha'_1(0), \dots, \alpha'_k(0)) = \sum_{j=1}^k \frac{d}{dt} \Big|_{t=0} \left(\prod_{1 \leq i < j} p_i \cdot \alpha_j(t) \cdot \prod_{j < i \leq k} p_i \right) \quad (2.1.1)$$

$$(dinv)_e(\alpha'(0)) = -\alpha'(0) \quad (2.1.2)$$

$$(dinv)_g(\alpha'(0)) = - (dR_{g^{-1}})_e \circ (dL_{g^{-1}})_g. \quad (2.1.3)$$

Proof. Recall that $T_{(p_1, \dots, p_k)} \prod_{i=1}^k G$ can be identified with $\prod_{i=1}^k T_{p_i} G$. To prove (2.1.1), it suffices to find the image of $(0, \dots, \alpha'_j(0), \dots, 0)$. To find it, consider the curve $(\alpha_i(t))_{i=1}^k$ such that for $i \neq j$, $\alpha_i(t) = p_i$, so that $\alpha'_i(0) = 0 \in T_{p_i} G$. Next, $m_k(\alpha_1(t), \dots, \alpha_k(t)) = \prod_{1 \leq i < j} p_i \cdot \alpha_j(t) \cdot \prod_{j < i \leq k} p_i$. Hence we get (2.1.1). In particular, $(dm)_{(e,e)}((u, v)) = u + v$. To get (2.1.2), just notice that $m \circ (inv, Id)$ is a constant map, $(d(inv, Id))_e(u) = ((dinv)_e(u), (dId)_e(u))$, and using the chain rule. From these, we obtain the result. For (2.1.3), we apply the chain rule to $inv = R_{g^{-1}} \circ inv \circ L_{g^{-1}}$ at the point g . $\square$

Later, we will see that working in a simply connected space is more important than working in a merely connected one. Then we may need a way to pull back a group structure via a covering map.

2.1.14 Proposition. Let G be a Lie group, H be a connected manifold and $\pi : H \rightarrow G$ a covering map⁵. For $e \in \pi^{-1}(e_G)$, there is a unique Lie group structure on H such that π is a Lie group morphism, e is the neutral element and the kernel of π is a subgroup of the center of H .

Proof. The task is to define the binary operation b in H and show that it works. H is path connected.

$$\begin{aligned} b: H \times H &\rightarrow H \\ (x, y) &\mapsto \omega(1), \end{aligned}$$

where ω is the lifting of $L_{\pi(x)} \circ \pi \circ \beta$, where β is a path starting from e_H to y .

Now suppose we choose another path β' starting from e_H to y . We are sure that $\beta' = \beta \star \alpha$, where α is a loop at y . Then ω' is the lifting of $L_{\pi(x)} \circ \pi \circ (\beta \star \alpha) = (L_{\pi(x)} \circ \pi \circ \beta) \star (L_{\pi(x)} \circ \pi \circ \alpha)$, that is $\omega \star (\text{lifting of } L_{\pi(x)} \circ \pi \circ \alpha \text{ at } \omega(1))$.⁶ So $\omega'(1) = \omega(1)$. Hence, b is well-defined.

Clearly e_H is a neutral element for b . We can also see that it is associative. For $x \in H$, we can also check that the endpoint of the lifting of $inv \circ \pi \circ \alpha$ is its inverse if α is the path joining e_H to x .

This construction is nothing else than the lifting of $m \circ (\pi \times \pi)$, which is smooth (because the lift of a smooth map through a local diffeomorphism is smooth) and unique.

Moreover, by construction of b , π is a group homomorphism. Then π is a Lie group morphism.

Let $a \in \ker(\pi)$ and $h \in H$, whose path α is the path starting from e to h . Since the kernel is normal, we can define a continuous path $\gamma : I \rightarrow \ker(\pi)$, $\gamma(t) = \alpha(t)a(\alpha(t))^{-1}$. This must be constant because the kernel (fiber) of a covering map is discrete. Hence $hah^{-1} = \gamma(1) = \gamma(0) = a$. Thus $ah = ha$. $\square$

2.1.15 Example (Universal Cover of $\mathbf{SL}(2, \mathbb{R})$). As the determinant map is continuous, $\mathbf{SL}(2, \mathbb{R})$ has to be connected. But the Iwasawa decomposition states every element $M \in \mathbf{SL}(2, \mathbb{R})$ is written uniquely

$$M = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \quad \text{with } \theta \in [-\pi, \pi] / \{-\pi \sim \pi\}, a > 0, u \in \mathbb{R}.$$

⁵Also smooth.

⁶This second lifting is a loop as α is a loop. To see it, $\pi(y)$ and $L_{\pi(x)}$ are joined by a path, let say ε . Then $(\pi \circ \alpha) \star \varepsilon = \varepsilon \star (L_{\pi(x)} \circ \pi \circ \alpha)$. Checking carefully the right starting point of the lifting, the lifting $\tilde{\varepsilon}$ of ε is also a loop.

This identification is smooth correspondence from $\mathbf{S} = S^1 \times \mathbb{R}_{>0} \times \mathbb{R}$ to $\mathbf{SL}(2, \mathbb{R})$. Conversely,

$$M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \in \mathbf{SL}(2, \mathbb{R}) \mapsto \left(\text{atan2}^7(m_{21}, m_{11}), \sqrt{m_{11}^2 + m_{21}^2}, \frac{m_{11}m_{12} + m_{21}m_{22}}{m_{11}^2 + m_{21}^2} \right) \in \mathbf{S}.$$

Then $\mathbf{SL}(2, \mathbb{R})$ is diffeomorphic to $\mathbf{S}$ which is not simply connected. Let call $f : \mathbf{S} \rightarrow \mathbf{SL}(2, \mathbb{R})$. This identification can gives us the universal cover of $\mathbf{SL}(n, \mathbb{R})$, $\tilde{\mathbf{S}} = \mathbb{R} \times \mathbb{R}_{>0} \times \mathbb{R}$. The covering map is

$$\begin{aligned} \pi : \mathbb{R} \times \mathbb{R}_{>0} \times \mathbb{R} &\longrightarrow \mathbf{SL}(2, \mathbb{R}) \\ (\theta, a, u) &\longmapsto \pi((\theta, a, u)) = \begin{pmatrix} a \cos \theta & au \cos \theta - a^{-1} \sin \theta \\ a \sin \theta & au \sin \theta + a^{-1} \cos \theta \end{pmatrix}. \end{aligned}$$

Proposition 2.1.14 says $\mathbb{R} \times \mathbb{R}_{>0} \times \mathbb{R}$ has an unique group structure to make it Lie group and π Lie group morphism if we fix $e = (0, 1, 0) \in \pi^{-1}(I_2)$ the identity. One can define lift the operation from $\mathbf{SL}(2, \mathbb{R})$, as in the proof. The deal is, there is group structure ω on $\mathbf{S}$ making $f : (\mathbf{S}, \omega) \rightarrow \mathbf{SL}(2, \mathbb{R})$ an isomorphism of group. Then, the unique group structure on $\tilde{\mathbf{S}}$ doing the job is the lift of ω .

About another concrete example, one can reconstruct, is the group law in $\mathbb{R}^n$, if we consider the covering map $\pi : \mathbb{R}^n \rightarrow T^n$. But choosing $(0, 0, 0) \in \pi^{-1}(e)$, we get the classic group operation in $\mathbb{R}^n$.

2.1.16 Example (Universal cover of $\mathbf{SO}(3, \mathbb{R})$). Let's start by defining the Pauli matrices

$$P_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad P_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathcal{M}_2(\mathbb{C}).$$

As they are linearly independent,

$$\begin{aligned} \mathbb{A} : \mathbb{R}^3 &\longrightarrow V = \text{span}_{\mathbb{C}}\{P_1, P_2, P_3\} \\ (x_1, x_2, x_3) &\longmapsto \mathbb{A}((x_1, x_2, x_3)) = x_1 P_1 + x_2 P_2 + x_3 P_3, \end{aligned}$$

is an isomorphism. Noticing $P_k^2 = I_2$ and $P_k P_l = -P_l P_k$ for $k \neq l$,

$$\text{Tr}(P_k P_l) = 2\delta_{kl} \implies \text{Tr}(A(x)A(y)) = \text{Tr}\left(\sum_{k,l} x_k y_l P_k P_l\right) = 2 \langle x, y \rangle.$$

So, $g : V^2 \rightarrow \mathbb{R}, (u, v) \mapsto \frac{1}{2} \text{Tr}(uv)$ is an inner product (metric) that makes $\mathbb{A}$ an isometry.

Next, for any $U \in \mathbf{SU}(2) = \{M \in \mathcal{M}_2(\mathbb{C}) : \det(M) = 1, U^\dagger U = I_2\}$, let's define $\bar{U} : V \rightarrow V, v \mapsto UvU^{-1}$. It is well-defined, because V is the set of all matrices X in $\mathcal{M}_2(\mathbb{C})$ that verifies $X^\dagger = -X$ and $\text{Tr}(X) = 0$. It is an isomorphism and $g(\bar{U}(v), \bar{U}(w)) = g(v, w)$. Then it is an isometry and $\mathbb{A}^{-1}\bar{U}\mathbb{A}$ is a linear isometry of $\mathbb{R}^3$. Then we have a map $\tilde{\phi}$ from $\mathbf{SU}(2)$ to $\mathbf{O}(3, \mathbb{R})$. One can check also, it is continuous and $\bar{I}_2 = I_3$ then the range of $\tilde{\phi}$ is subset of $\mathbf{SO}(3, \mathbb{R})$. Hence, we have

$$\begin{aligned} \phi : \mathbf{SU}(2) &\longrightarrow \mathbf{SO}(3, \mathbb{R}) \\ U &\longmapsto \phi(U) = \tilde{U}. \end{aligned}$$

Recall that $\mathbf{SU}(2)$ is not a complex Lie group but it is a real Lie group. His real structure is the universal cover of $\mathbf{SO}(3, \mathbb{R})$. We just need to see $\mathbf{SU}(2)$ is simply connected and ϕ is covering map, the group

⁷atan2(y, x) is angle between x-axis and $\vec{OP}$, for $P = \begin{pmatrix} x \\ y \end{pmatrix}$ continuous as we identify $-\pi \sim \pi$ and defined as $\begin{pmatrix} m_{11} \\ m_{21} \end{pmatrix}$ is nonzero.

operation in $\mathbf{SU}(2)$ coincides with the one we construct in Proposition 2.1.14. Being group morphism is straightforward as

$$\overline{U_1 U_2}(v) = \overline{U_1}(\overline{U_2}(v)) \implies \phi(U_1 U_2) = \mathbb{A}^{-1} \overline{U_1 U_2} \mathbb{A} = (\mathbb{A}^{-1} \overline{U_1} \mathbb{A})(\mathbb{A}^{-1} \overline{U_2} \mathbb{A}) = \phi(U_1) \phi(U_2).$$

Also, $U^\dagger U = I_2$ and $\det(U) = 1$ means $U = \begin{pmatrix} a+ib & c+id \\ -c+id & a-ib \end{pmatrix}$, $a^2 + b^2 + c^2 + d^2 = 1$ for some $a, b, c, d \in \mathbb{R}$. Then $\mathbf{SU}(2)$ is homeomorphic to S^3 , which is simply connected. We can prove that it is covering map but I think it is heavy computation by the known formula, using quaternion group: Euler–Rodrigues formula. We will not present it this way but use an important theorem to see it.

2.1.17 Theorem. Let $\pi : G \longrightarrow H$ be a morphism of connected Lie groups. π is a covering map if and only if $(d\pi)_a$ is an isomorphism, for all $a \in G$.

Proposition 2.1.6 tells us that $(d\pi)_a$ is an isomorphism, for all $a \in G$ is equivalent $(d\pi)_a$ being an isomorphism, for for some $a \in G$.

Proof. If π is covering map, then it is local diffeomorphism. So, $(d\pi)_a$ is an isomorphism. Conversely, assume $(d\pi)_a$ is isomorphism, for any $a \in G$. By the Inverse Function Theorem, there is V and W neighbourhoods of identity of G and of H such that π restricts on V is diffeomorphism onto W . By connectedness, each element $y \in H$ can be expressed using the elements in W , then y is in the range of π . So π is surjective. For any $y \in H$, $\pi^{-1}(y)$ is discrete because otherwise, there $x \in \pi^{-1}(y)$ such that π is constant (y) near to x , which contradicts the fact locally diffeomorphism. Let's call $J = \text{Ker}(\pi)$, discrete set. For any $y \in H$ and $x \in \pi^{-1}(y)$,

$$a \in \pi^{-1}(y) \iff \pi(a) = \pi(y) \iff x^{-1}a \in \text{Ker}(\pi) \text{ then } \pi^{-1}(y) = xJ.$$

Also there are V_x and W_y open neighbourhoods of x and y such that $\pi : V_x \longrightarrow W_y$ is diffeomorphism. Clearly, for any $a \in J$, aV_x is subset of $\pi^{-1}(W_y)$. If $g \in \pi^{-1}(W_y)$, $\pi(g) \in W_y$, then $\pi(v) = \pi(g)$ for some $v \in V_x$. So $a := gv^{-1} \in \text{Ker} \pi$ and $g \in aV_x$. Hence $\pi^{-1}(y) = \bigsqcup_{a \in J} aV_x$. The disjoint union is coming from the fact if there is $g \in aV_x \cap bV_x$ namely $g = au = bv$ for some $u, v \in V_x$, $\pi(u) = \pi(a)\pi(u) = \pi(b)\pi(v) = \pi(v)$, then $a = b$. Finally, for any $a \in J : \pi : aV_x \longrightarrow W_y$ are also diffeomorphism as composition of $L_{a^{-1}}$ and $\pi : V_x \longrightarrow W_y$. $\square$

2.1.18 Example (Continue). This theorem allows us to complete Example 2.1.16. We have proven that π is a Lie group morphism. We claim first that $T_e \mathbf{SO}(3, \mathbb{R}) = \{M \in \mathcal{M}_3(\mathbb{R}) : M^T = -M\}$. Indeed if $M^T = -M$. $(e^{tM})^T(e^{tM}) = e^{tM^T + tM} = e^0 = I_3$. As, $\det(e^{tM}) > 0$, then $\gamma(t) = e^{tM}$ is a curve on $\mathbf{SO}(3, \mathbb{R})$, as $\det(e^{tM}) = e^{\text{Tr}(tM)} > 0$ starting at I_3 . Then $M = \gamma'(0) \in T_e \mathbf{SO}(3)$. Conversely if $M \in T_e \mathbf{SO}(3, \mathbb{R})$, there is a curve $\gamma(t) \in \mathbf{SO}(3, \mathbb{R})$ such that, starting at I_3 and $\gamma'(0) = M$. One can differentiate $\gamma(t)^T \gamma(t) = I_3$, using $\frac{d}{dt} \Big|_{t=0} \gamma(t)^T = \left(\frac{d}{dt} \Big|_{t=0} \gamma(t) \right)^T$ with the product rule in Lemma 2.1.13, in clever way, we get

$$M + M^T = \gamma'(0)\gamma(0)^T + \gamma(0)^T \gamma'(0)^T = 0 \implies M^T = -M.$$

We do the same reasoning to get $T_e \mathbf{SU}(2) = \{X \in \mathcal{M}_2(\mathbb{C}) : X^\dagger = -X, \det(X) = 1\}$. Such thing we will generalize later but for now, we compute them directly. On another hand, we can write directly

$$\phi(U)(x) = \mathbb{A}^{-1}(U \mathbb{A}(x) U^{-1}) \implies ((d\phi)_e(M))(x) = \frac{d}{dt} \Big|_{t=0} \mathbb{A}^{-1}(e^{tM} \mathbb{A}(x) e^{-tM}) = \mathbb{A}^{-1} \left(\frac{d}{dt} \Big|_{t=0} e^{tM} \mathbb{A}(x) e^{-tM} \right).$$

Then $((d\phi)_e(M))(x) = \mathbb{A}^{-1}(M\mathbb{A}(x) - \mathbb{A}(x)M)$. It is clear that $\{iP_k : k = 1, 2, 3\}$ forms a basis of $T_e\mathbf{SU}(2)$. One can compute $P_k P_l - P_l P_k = 2i\varepsilon_{klm}^8 P_m$ for $m \neq k, l$. Then, by linearity

$$((d\phi)_e(iP_k))(x) = \sum_{l,m \neq k,l} 2x_l \mathbb{A}^{-1}(i^2 \varepsilon_{klm} P_m) = 2 \sum_{l,m \neq k,l} \varepsilon_{lkm} x_l e_m = 2x \wedge e_k. \quad (2.1.4)$$

$$\implies (d\phi)_e(iP_k) = N_k, \quad N_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{pmatrix}, \quad N_2 = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}, \quad N_3 = \begin{pmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (2.1.5)$$

Clearly the (N_k) is basis of $T_e\mathbf{SO}(3, \mathbb{R})$. Then $(d\phi)_e$ is an isomorphism and ϕ is a covering map.

Proposition 2.1.14 provide us further examples of Lie groups. We have another way, somehow its inverse, assume if we have Γ subgroup of Lie group H , when the quotient H/Γ nice to be Lie group.

2.1.19 Proposition. Let H be Lie group of center $Z(H)$ and Γ be a discrete subgroup of $Z(H)$. Then H/Γ has unique group structure making it to be Lie group and the quotient covering map.

The idea of proof considering the action of Γ on H , $\gamma \cdot h = h\gamma$. This action is free and proper so the quotient manifold Theorem 1.4.1 is applicable.

Note that if Γ is just normal subgroup and closed, we can still have a Lie group structure on the quotient H/Γ make the quotient Lie group morphism. It is by one fact that I skip, about Homogeneous Space Characterization Theorem, found in Lee (2003) (Theorem 21.18).

2.2 Representation of Lie groups

For group G and vector space V , a representation of G is a group homomorphism $\rho : G \longrightarrow \mathbf{GL}(V)$. Now, in our case the group has more structure then we requires also smoothness of it to be representation of Lie group. A subspace $W \subseteq V$ is called G -invariant or subrepresentation if $\rho(g)W \subseteq W$ for all $g \in G$. The representation is irreducible if its only subrepresentations are $\{0\}$ and V . A morphism between representations (V, ρ) and (W, ρ') is a linear map $T : V \rightarrow W$ such that $T\rho(g) = \rho'(g)T$ for all g . The classical backgrounds and constructions (sum, product, tensor, etc) can be seen in some of our references Hall (2013), Fulton and Harris (2013). We just underline the Schur's lemma, for V, W are irreducible representations, any morphism $T : V \rightarrow W$ is either zero or an isomorphism.

2.2.1 Adjoint representation of Lie group

Proposition 2.1.10 tells that any neighbourhood of identity of a connected Lie group can determine entirely the Lie group. This leads us to study the tangent space at the identity.

Firstly, let ψ_g be the inner automorphism of G , $\psi_g(h) = ghg^{-1}$. Next, Ad is define as

$$\begin{aligned} \text{Ad} : G &\longrightarrow \mathbf{GL}(T_e G) \\ g &\longmapsto \text{Ad}(g) = (d\psi_g)_e. \end{aligned}$$

The fact that $\psi_{gh} = \psi_g \circ \psi_h$ and the functoriality of differentiation ensures us that is a representation of G on its tangent at identity. This is called the adjoint representation of G .

2.2.2 Lemma. Let $\rho : G \longrightarrow H$ a Lie group morphism. Then

- (a) ρ respects the conjugation action i.e. $\rho \circ \psi_g = \psi_{\rho(g)} \circ \rho$;

⁸This is the Levi-Civita symbol which 0 if two of index are equal and signature of the permutaton (klm) otherwise.

(b) ρ respects the adjoint action *i.e.* $(d\rho)_e \circ \text{Ad}(g) = (\text{Ad}(\rho(g)) \circ (d\rho)_e$.

Proof. (a) is straightforward and we use it with the functoriality of the differentiation to get (b). $\square$

We want to see that a Lie group morphism whose domain is connected is uniquely determined by its differential at the identity. To get it, let's define again

$$\text{ad} = (d\text{Ad})_e : T_e G \rightarrow \text{End}(T_e G) \text{ and } L(X, Y) := \text{ad}(X)(Y). \quad (2.2.1)$$

2.2.3 Proposition. The $L : T_e G \times T_e G \rightarrow T_e G$, defined at (2.2.1) is skew-symmetric bilinear map and verifies the Jacobi identity⁹. Moreover if $\rho : G \rightarrow H$ is Lie group morphism, then it respects the adjoint action $L = \text{ad}$, that is $(d\rho)_e(\text{ad}(X)(Y)) = \text{ad}((d\rho)_e(X))((d\rho)_e(Y))$.

Proof. Let's go step by step. First

$$\begin{aligned} \text{Ad}(g)(\beta'(0)) &= \left. \frac{d}{dt} \right|_{t=0} g\beta(t)g^{-1} \in T_e G \\ \implies \text{ad}(\alpha'(0))(\beta'(0)) &= \left. \frac{d}{ds} \right|_{s=0} \left. \frac{d}{dt} \right|_{t=0} \alpha(s)\beta(t)\alpha(s)^{-1} \in T_e(T_e G) \cong T_e G \\ \implies \text{Ad}(h)(\text{ad}(\alpha'(0))(\beta'(0))) &= \left. \frac{d}{ds} \right|_{s=0} \left. \frac{d}{dt} \right|_{t=0} h\alpha(s)\beta(t)\alpha(s)^{-1}h^{-1} \in T_e G \\ \implies \text{ad}(\gamma'(0))(\text{ad}(\alpha'(0))(\beta'(0))) &= \left. \frac{d}{dr} \right|_{r=0} \left. \frac{d}{ds} \right|_{s=0} \left. \frac{d}{dt} \right|_{t=0} \gamma(r)\alpha(s)\beta(t)\alpha(s)^{-1}\gamma(r)^{-1} \in T_e(T_e G) \cong T_e G. \end{aligned} \quad (2.2.2) \quad (2.2.3)$$

Let's see them as derivation. Consider the curves (by abuse of vocabulary) $(I_\varepsilon = (-\varepsilon, \varepsilon))$,

$$\begin{aligned} \delta : I_\varepsilon \times I_\varepsilon &\longrightarrow G, & \omega : I_\varepsilon \times I_\varepsilon \times I_\varepsilon &\longrightarrow G \\ (s, t) &\mapsto \alpha(s)\beta(t)\alpha(s)^{-1}, & (r, s, t) &\mapsto \gamma(r)\alpha(s)\beta(t)\alpha(s)^{-1}\gamma(r)^{-1}. \end{aligned}$$

Let $f \in \mathcal{C}^\infty(G)$, we are interesting in what is $(\text{ad}(\alpha'(0))(\beta'(0)))f$.

$$\begin{aligned} (\text{ad}(\alpha'(0))(\beta'(0)))f &= \left. \frac{d}{ds} \right|_{s=0} \left. \frac{d}{dt} \right|_{t=0} f \circ \delta(s, t) = \left. \frac{d}{dt} \right|_{t=0} \left. \frac{d}{ds} \right|_{s=0} f \circ \delta(s, t) \\ &= \left. \frac{d}{dt} \right|_{t=0} F(t) \quad \text{if } F(t) = \left. \frac{d}{ds} \right|_{s=0} f \circ \delta(s, t) = (d(f \circ \delta(-, t)))_0 \\ &= \left. \frac{d}{dt} \right|_{t=0} \left((d(f \circ m_3 \circ (\alpha, c_{\beta(t)}, \text{inv} \circ \alpha)))_0 \right), \quad \text{where } c_{\beta(t)} \text{ is constant map} \\ &= \left. \frac{d}{dt} \right|_{t=0} \left((df)_{\beta(t)} \circ (dm_3)_{(e, \beta(t), e)}(\alpha'(0), 0, -\alpha'(0)) \right) \text{ by (2.1.2)} \\ &= \left. \frac{d}{dt} \right|_{t=0} \left((df)_{\beta(t)}(\alpha'(0)\beta(t) - \beta(t)\alpha'(0)) \right) \text{ by (2.1.1)} \\ &= \left. \frac{d}{dt} \right|_{t=0} \left. \frac{d}{ds} \right|_{s=0} (f(\alpha(s)\beta(t)) - f(\beta(t)\alpha(s))). \end{aligned} \quad (2.2.4)$$

⁹see the definition (2.3.1)

This shows the skew symmetry. To see the Jacobi Identity, we do the similar machinery to get

$$\begin{aligned} \left(\text{ad}(\gamma'(0))(\text{ad}(\alpha'(0))(\beta'(0))) \right) f &= \frac{d}{dt} \Big|_{t=0} \frac{d}{ds} \Big|_{s=0} \frac{d}{dr} \Big|_{r=0} \left(f(\gamma(r)\alpha(s)\beta(t)) - f(\gamma(r)\beta(t)\alpha(s)) \right. \\ &\quad \left. + f(\beta(t)\alpha(s)\gamma(r)) - f(\alpha(s)\beta(t)\gamma(r)) \right). \end{aligned}$$

We can compute now and get

$$\text{ad}(\gamma'(0))(\text{ad}(\alpha'(0))(\beta'(0)))f + \text{ad}(\alpha'(0))(\text{ad}(\beta'(0))(\gamma'(0)))f + \text{ad}(\beta'(0))(\text{ad}(\gamma'(0))(\alpha'(0)))f = 0.$$

Now, let $\rho : G \rightarrow H$ Lie group morphism. Equation (2.2.2) gives us

$$\begin{aligned} (\text{d}\rho)_e(\text{ad}(\alpha'(0))(\beta'(0))) &= \frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} \rho(\alpha(s)\beta(t)\alpha(s)^{-1}) \in T_e H \\ &= \frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} \left((\rho(\alpha(s))) (\rho(\beta(t))) (\rho(\alpha(s)))^{-1} \right) \\ &= \text{ad} \left(\frac{d}{ds} \Big|_{s=0} \rho(\alpha(s)) \right) \left(\frac{d}{dt} \Big|_{t=0} \rho(\beta(t)) \right) \\ &= (\text{d}\rho)_e(\text{ad}(\alpha'(0))) (\text{d}\rho)_e(\text{ad}(\beta'(0))). \end{aligned} \tag{2.2.5}$$

Hence ρ respects adjoint action. One can get it directly from Lemma 2.2.2 (b). $\square$

2.2.4 Example. Let G is linear Lie group. For fixed A in G , $\psi_A(M) = AMA^{-1}$. Identifying $T_e G$ by a linear subspace of $T_e \mathbf{GL}(n, \mathbb{R}) = \mathcal{M}_n(\mathbb{R})$, for any $A \in G$ and $X \in T_e G$:

$$\text{Ad}(A)(X) = \frac{d}{dt} \Big|_{t=0} A \exp(tX) A^{-1} = \frac{d}{dt} \Big|_{t=0} AX \exp(tX) A^{-1} = AXA^{-1}.$$

Now, for $Y \in T_e G$:

$$\begin{aligned} \text{ad}(Y)(X) &= \frac{d}{dt} \Big|_{t=0} (\exp(tY)X \exp(-tY)) = \left(\left(\frac{d}{dt} \exp(tY) \right) X \exp(-tY) + \exp(tY) X \left(\frac{d}{dt} \exp(-tY) \right) \right) \Big|_{t=0} \\ &= YX - XY. \end{aligned}$$

2.3 Lie algebras

2.3.1 Definition (Lie Algebra). Let $\mathbb{K}$ be field. A Lie Algebra over $\mathbb{K}$ is a vector space V over $\mathbb{K}$ together with a Lie bracket $[-, -]$, namely a skew symmetric $\mathbb{K}$ -bilinear which satisfies the Jacobi identity

$$\forall X, Y, Z \in V : [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0. \tag{2.3.1}$$

Now, the Lie algebra morphism is linear map between Lie algebra that respects the Lie bracket.

2.3.2 Example. Proposition 2.2.3 shows us that the $T_e G$ with ad is Lie algebra and any Lie group morphism induces a Lie algebra morphism by taking its differential.

2.3.3 Definition. A Lie subalgebra $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is a linear subspace that is closed under the Lie bracket. But an ideal $\mathfrak{k}$ of $\mathfrak{g}$ is a subalgebra such that for any $X \in \mathfrak{k}$ and $Y \in \mathfrak{g}$, $[X, Y] \in \mathfrak{k}$.

2.3.4 Example. It is straightforward to see that the set of vector fields on manifold M , $\mathfrak{X}(M)$ is vector space. For $X, Y \in \mathfrak{X}(M)$, $X \circ Y$ may not satisfy the product rule but $X \circ Y - Y \circ X$ is vector field in M . It is not hard to check that $[X, Y] := X \circ Y - Y \circ X$ is a Lie bracket on $\mathfrak{X}(M)$. Thus we get a lie algebra. Now let's assume that $G = M$ is a Lie group. We mean by a left-invariant vector, vector field $X \in \mathfrak{X}(G)$ such that: for any $g, h \in G$ $(dL_g)_h(X_h) = X_{gh}$ (or simply $(dL_g)(X) = X$). Again, it is clear that the set of vector fields which are left-invariant $\mathfrak{L}(G)$ form a linear subspace of $\mathfrak{X}(G)$. Moreover, for $f \in \mathcal{C}^\infty(G)$ and $X, Y \in \mathfrak{L}(G)$,

$$\begin{aligned} (dL_g)_h([X, Y]_h)f &= ([X, Y](f \circ L_g))(h) = (X(Y(f \circ L_g)) - Y(X(f \circ L_g)))(h) \\ &= (X((dL_g)_-(Y_-)))(f)_{gh} - (Y((dL_g)_-(X_-)))(f)_{gh} \\ &= (X(Y(f)))(gh) - (Y(X(f)))(gh) = [X, Y]_{gh}f. \end{aligned}$$

Thus $[X, Y]$ is also left-invariant. Hence $\mathfrak{L}(G)$ is Lie subalgebra of $\mathfrak{X}(G)$.

Now, there are two Lie algebra from a Lie group. We want to see that they are the same thing.

2.3.5 Lemma. For a Lie group G and $v = \alpha'(0) \in T_e G$, $X = X^{(v)}$ defined bellow in (2.3.2), is the unique left-invariant vector field on G such that $X_e = v$.

$$\begin{aligned} X(f): G &\longrightarrow \mathbb{R} \\ g &\longmapsto X(f)(g) = \left. \frac{d}{dt} \right|_{t=0} f(g\alpha(t)). \end{aligned} \quad (2.3.2)$$

In other words,

$$X(g) := (dL_g)_e(v).$$

Proof. Clearly $X_e = v$. Let $g, h \in G$ and $f \in \mathcal{C}^\infty(G)$.

$$((dL_g)_h(X_h))f = X_h(f \circ L_g) = (X(f \circ L_g))(h) = \left. \frac{d}{dt} \right|_{t=0} f(gh\alpha(t)) = X(f)(gh) = X_{gh}f.$$

Then X is left-invariant. Now let $Y \in \mathfrak{L}(G)$ such that $Y_e = v = X_e$. Take arbitrary $g \in G$.

$$Y_g = Y_{ge} = ((dL_g)_e(Y_e)) = ((dL_g)_e(X_e)) = X_{ge} = X_g.$$

Hence a left-invariant vector field is uniquely determined by its value at identity. □

So far, this lemma tells us that, as set, $T_e G$ and $\mathfrak{L}(G)$ is equivalent. Moreover,

2.3.6 Proposition. The correspondence, constructed in Lemma 2.3.5

$$\begin{aligned} \mathcal{F}: (T_e G, ad) &\longrightarrow (\mathfrak{L}(G), [-, -]) \\ v = \alpha'(0) &\longmapsto \mathcal{F}(v = \alpha'(0)) = X^{(v)}, \end{aligned}$$

give us an identification (isomorphism of Lie algebra) of $T_e G$ and $\mathfrak{L}(G)$.

Proof. The bijection is trivial. We use the uniqueness in Lemma 2.3.5 to get the linearity. Now, let $v = \alpha'(0)$, $w = \beta'(0)$ and $X = \mathcal{F}(v)$, $Y = \mathcal{F}(w)$. Using (2.3.2), for any $f \in \mathcal{C}^\infty(G)$ we have

$$\begin{aligned} [X, Y]_e f &= (X(Y(f)) - Y(X(f)))(e) = \left(\frac{d}{dt} \Big|_{t=0} Y(f)(\alpha(t)) \right) - \left(\frac{d}{ds} \Big|_{s=0} X(f)(\beta(s)) \right) \\ &= \left(\frac{d}{dt} \Big|_{t=0} \frac{d}{ds} \Big|_{s=0} f(\alpha(t)\beta(s)) \right) - \left(\frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} f(\beta(s)\alpha(t)) \right) \\ &= (\text{ad}(v)(w))f. \quad \text{by (2.2.4).} \end{aligned}$$

Hence $\mathcal{F}(\text{ad}(v)(w)) = [\mathcal{F}(v), \mathcal{F}(w)]$. □

The composition of Lie algebra morphisms is clearly Lie algebra morphism. Then we can talk about the isomorphism of Lie algebras and category of Lie algebras $\mathcal{LA}$. For example, Proposition 2.3.6 gives the isomorphism between $(T_e G, \text{ad})$ and $(\mathfrak{L}(G), [-, -])$. On another hand, we have established a functor (it respects clearly composition and identity)

$$\begin{aligned} \mathbb{T} : \mathcal{LG} &\rightarrow \mathcal{LA} & (2.3.3) \\ G &\mapsto \mathbb{T}(G) = (T_e G, \text{ad}) \\ G &\xrightarrow{\rho} H \mapsto T_e \xrightarrow{(\text{d}\rho)_e} T_e H. \end{aligned}$$

2.3.7 Remark (Product). Given Lie algebra $\mathfrak{g}$ and $\mathfrak{h}$, one can define a bracket on $\mathfrak{g} \times \mathfrak{h}$ by

$$[(X_1, X_2), (Y_1, Y_2)] = ([X_1, Y_1], [X_2, Y_2]).$$

Clearly, it is a Lie bracket and makes the projections Lie algebra morphisms. Moreover, it is straightforward to see that for any pair of Lie algebra morphisms $g : \mathfrak{a} \rightarrow \mathfrak{g}$ and $h : \mathfrak{a} \rightarrow \mathfrak{h}$, the unique map that commutes with the projections $g \times h$ is also a Lie algebra morphism. Hence, it is the abstract product of Lie algebras.

The functor $\mathbb{T}$ respects products because $T_{(e,e)}(G \times H) = T_e G \times T_e H$ and for any $\alpha'(0) = (\alpha'_1(0), \alpha'_2(0))$ and $\beta'(0) = (\beta'_1(0), \beta'_2(0))$ in $T_{(e,e)}(G \times H)$, using (2.2.2),

$$\begin{aligned} [\alpha'(0), \beta'(0)] &= \frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} (\alpha_1(s), \alpha_2(s))(\beta_1(t), \beta_2(t))(\alpha_1(s), \alpha_2(s))^{-1} \\ &= \left(\frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} \alpha_1(s)\beta_1(t)\alpha_1(s)^{-1}, \frac{d}{ds} \Big|_{s=0} \frac{d}{dt} \Big|_{t=0} \alpha_2(s)\beta_2(t)\alpha_2(s)^{-1} \right) \\ &= ([\alpha'_1(0), \beta'_1(0)], [\alpha'_2(0), \beta'_2(0)]). \end{aligned}$$

2.4 Correspondence between Lie groups and Lie algebras

So far, we have the functor $\mathbb{T}$, defined at (2.3.3) from the category of Lie groups to the category of Lie algebras. Also, it is straightforward to see that Lie subgroups correspond to Lie subalgebras by considering the inclusion and differentiating it, giving an injective Lie algebra morphism.

2.4.1 Proposition. If H is a Lie subgroup of G , then $\mathfrak{h} = T_e H$ is a Lie subalgebra of $\mathfrak{g}$.

We also want to see the converse of this fact: given a Lie algebra, is there a Lie group that corresponds to it? Given a Lie algebra $\mathfrak{g}$ coming from a Lie group G , what information can we get about G from

$\mathfrak{g}$? We will see that there is a correspondence between Lie subalgebras and Lie groups, under a small adjustment.

This objective provides a more convincing definition of a Lie subgroup. We know that $T_e(S^1 \times S^1) = \mathbb{R}^2$ and it is straightforward to see that the bracket from an abelian group is trivial. Then the subalgebras are all the lines passing through the origin. One can show, in the same way we did with Example 2.1.8, that only those with rational slope are embedded. Then, the correct definition of Lie subgroup should be the immersed one. So, why have we fixed this? If we say only Lie subgroup, we mean immersed.

This part is just motivation, but we need to study the exponential map to have tools about the completion of the correspondence.

A representation of a Lie group is a Lie group morphism from G to $\mathbf{GL}(V)$. This induces a Lie algebra morphism from $T_e G$ to $\mathfrak{gl}(V)$, which we call a representation of $T_e G$. More generally,

2.4.2 Definition. For a Lie algebra $\mathfrak{g}$ and a vector space V , a representation of $\mathfrak{g}$ is a Lie algebra morphism $\varphi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$.

2.4.3 Proposition. Let (ρ, V) be a representation of a Lie group G and (φ, V) the induced representation of the Lie algebra $T_e G$. Then (ρ, V) is irreducible if and only if (φ, V) is irreducible.

3. Exponential map

In this chapter, we construct the exponential map on a Lie group and study its properties. This is totally algebraic definition, without choosing a metric. We are interested in situations where this exponential map coincides with the exponential map in a general Riemannian manifold. This occurs in certain cases where the Lie group admits a bi-invariant metric. For left-invariant metrics only, the situation remains interesting, since one can define Cartan connections which, as we will see, give rise to many important classes of curvature.

3.1 Exponential map and its first consequences

3.1.1 Definition. Let G be a Lie group. A one-parameter subgroup of G is a Lie group morphism $\gamma : \mathbb{R} \longrightarrow G$.

First, we recall that for any vector field X on a manifold M , $p \in M$ and $v \in T_p M$, there is a unique smooth curves $\alpha : I_\varepsilon = (-\varepsilon, \varepsilon) \longrightarrow M$ such that

$$\alpha(0) = p; \quad \alpha'(0) = v \quad \text{and} \quad X(\alpha(t)) = \alpha'(t).$$

Namely, α is an integral curve of X starting at p with initial velocity v .

Also, another fact is that such a path can be extended smoothly in all $t \in \mathbb{R}$ if X is left-invariant vector field on Lie group G .

3.1.2 Lemma. Let $\gamma : I \longrightarrow G$ be a smooth curve such that $0 \in I$ and $\gamma(0) = e$. γ is a one-parameter subgroup if and only if γ is an integral (maximal) curve of a vector left-invariant field.

Proof. Let γ be one-parameter subgroup. We consider $X(g) = (dL_g)_e \gamma'(0)$. Then it is a left-invariant and $X(\gamma(t)) = (dL_{\gamma(t)})_e \gamma'(0) = \gamma'(t)$. Conversely, assume that $\gamma : I \longrightarrow G$ is an integral curve of a left-invariant vector field X . We shall first prove that $\gamma(s+t) = \gamma(s)\gamma(t)$. To show it, let fix s , and consider $\alpha(t) = \gamma(s+t)$ and $\beta(t) = \gamma(s)\gamma(t)$. Clearly $\alpha(0) = \beta(0) = \gamma(s)$; $X(\alpha(t)) = \alpha'(t)$ and $\beta'(t) = (dL_{\gamma(s)})_{\gamma(t)} X(\gamma(t)) = X(\gamma(s)\gamma(t))$. The uniqueness of integral curve starting at $\gamma(s)$, of same velocity, gives us the equality $\alpha = \beta$. Now if we assume $I = (a, b)$ ($I = (a, b], \dots$). Take $s = b - \varepsilon \in I$ for very small ε and we can define $\gamma(s + 2\varepsilon) = \gamma(s)\gamma(2\varepsilon)$. In particular, we can extend γ to have I unbounded above. The same reasoning to see that I is unbounded below. By maximality, $I = \mathbb{R}$. $\square$

This lemma implies for any $v \in T_e G$ there is unique one-parameter subgroup γ_v such that $\gamma'_v(0) = v$. This important fact leads us the definition of exponential map.

3.1.3 Definition. The exponential map on a Lie group G is defined as

$$\begin{aligned} \exp = \exp_G : \mathfrak{g} &\longrightarrow G \\ v &\longmapsto \exp = \exp_G(v) = \gamma_v(1). \end{aligned}$$

3.1.4 Proposition (Naturality of the exponential map). For any $\rho : G \longrightarrow H$, Lie group morphism,

$$\exp_H \circ (d\rho)_e = \rho \circ \exp_G.$$

Proof. We note that, for $v \in T_e G$, $\rho \circ \gamma_v$ is a one-parameter subgroup of H with initial velocity $(d\rho)_e(v)$. By uniqueness $\gamma_{(d\rho)_e(v)} = \rho \circ \gamma_v$. Thus the result. $\square$

Also, one can get is

$$(\mathrm{d} \exp)_0(u) = \left. \frac{d}{dt} \right|_{t=0} \gamma_{tu}(1) = \left. \frac{d}{dt} \right|_{t=0} \gamma_u(t) = \gamma'_u(0) = u.$$

That means $(\mathrm{d} \exp)_0$ is identity, then isomorphism. So, there are open set, V of $\mathfrak{g}$ containing 0 and W for G where $\exp_G : V \rightarrow W$ is diffeomorphism.

3.1.5 Corollary. If G is connected, any Lie group morphism $\rho : G \rightarrow H$ is uniquely determined by $(\mathrm{d}\rho)_e$.

Proof. Since there are open sets V of $\mathfrak{g}$ containing 0 and W for G where $\exp_G : V \rightarrow W$ is diffeomorphism. The naturality of the exponential map gives uniquely the value of ρ on W from $(\mathrm{d}\rho)_e$. Hence, Proposition 2.1.10 gives us the conclusion. $\square$

Next, let's try to see some obvious fact about exponential map. For any $v \in T_e G$ and $\lambda \in \mathbb{R}$, $\left. \frac{d}{dt} \right|_{t=0} \gamma_v(\lambda t) = \lambda v$. By uniqueness, $\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$. So, $t \mapsto \exp(tv)$ is a one-parameter subgroup of G , for any $v \in T_e G$. Conversely if $\mathbb{E} : T_e G \rightarrow G$ smooth map such that differential at 0 is identity and for any $v \in T_e G$, $\mathbb{E}_v, t \mapsto \mathbb{E}(tv)$ is one-parameter subgroup. We have $\mathbb{E}'_v(0) = v$ and

$$\exp_G(v) = \mathbb{E}_v(1) = \mathbb{E}(v).$$

3.1.6 Example. Recall that for any $M \in \mathcal{M}_n(\mathbb{R})$ ¹, $\|M^k\|_2 \leq \|M\|_2^k$. Then the norm $e^M = \sum_{k \geq 0} \frac{1}{k!} M^k$ is bounded by $e^{\|M\|_2}$, that tell us the well-defined definition of exponential of matrices.

We know also that it verifies $\det(e^A) = e^{\mathrm{Tr}(A)} \neq 0$. Then $e^M \in \mathbf{GL}(n, \mathbb{R})$. Considering it as map from $T_e \mathbf{GL}(n, \mathbb{R}) = \mathbf{M}_n(\mathbb{R})$ to $\mathbf{GL}(n, \mathbb{R})$, it satisfies differential at 0, identity and for any $M \in \mathbf{M}_n(\mathbb{R})$, $t \mapsto e^{tM}$ is a one-parameter subgroup. So, the exponential map for $\mathbf{GL}(n, \mathbb{R})$ is just the exponential of matrices. Same reasoning is valid for all linear Lie groups.

3.1.7 Proposition. Let $F : G \rightarrow H$ be a group morphism between Lie groups. Continuity implies smoothness.

Proof. Let $\gamma : \mathbb{R} \rightarrow H$ be a group morphism and continuous. Let W be the open neighbourhood of 0 in $T_e H$ that makes the restriction of $\exp_H$ diffeomorphism. There is V open neighbourhood of 0 such that $3V \subseteq W$. Also there is $t_0 \in \mathbb{R}^*$ such that $\gamma([-t_0, t_0]) \subseteq \exp_H(V)$. Then $\gamma(t_0) = \exp_H(x_0)$ for some $x_0 \in V$. As $\exp_H$ is diffeomorphism, $\exp_H^{-1}$ exists on $2\exp_H(V)$,

$$\begin{aligned} \exp_H(2\exp_H^{-1}(\gamma(\frac{t_0}{2}))) &= \gamma(t_0) = \exp(x_0) \implies \gamma(\frac{t_0}{2}) = \exp(\frac{x_0}{2}) \in \exp(V) \\ \implies \exp_H(2\exp_H^{-1}(\gamma(\frac{1}{2}\frac{t_0}{2}))) &= \gamma(\frac{t_0}{2}) = \exp(\frac{x_0}{2}) \implies \gamma(\frac{t_0}{4}) = \exp(\frac{x_0}{4}) \in \exp(V) \\ \implies \gamma(\frac{t_0}{2^n}) &= \exp(\frac{x_0}{2^n}) \quad \forall n \in \mathbb{N} \\ \implies \gamma(\frac{m}{2^n}t_0) &= (\gamma(\frac{t_0}{2^n}))^m = (\exp(\frac{x_0}{2^n}))^m = \exp(\frac{m}{2^n}x_0) \quad \forall m \in \mathbb{N} \\ \implies \gamma(st_0) &= \exp(sx_0) \quad \forall s \in \mathbb{R} \quad \text{by density of } (\frac{m}{2^n})_{m,n} \in \mathbb{R} \\ \implies \gamma(t) &= \exp(\frac{x_0}{t_0}t) \quad \forall t \in \mathbb{R}. \end{aligned}$$

¹Similarly for $\mathcal{M}_n(\mathbb{C})$.

Hence γ is smooth. Now let $(x_i)_{i=1}^{i=d}$ be a basis of $T_e G$ and define a smooth map,

$$\begin{aligned} \varepsilon: \mathbb{R}^n &\longrightarrow G \\ u &\longmapsto \varepsilon(u) = \exp_G(u_1 x_1) \exp_G(u_2 x_2) \cdots \exp_G(u_n x_n). \end{aligned}$$

One can compute $(d\varepsilon)_0(u) = \left. \frac{d}{dt} \right|_{t=0} \varepsilon(tu) = u_1 X_1 + u_2 X_2 + \cdots + u_n X_n$. Then $(d\varepsilon)_0$ is isomorphism. Then there is U and V neighbourhoods of 0 and e where ε is an diffeomorphism. On the other hand $F \circ \varepsilon = F_1 F_2 \cdots F_n$, where each $F_k(u) = F(\exp(u_k x_k))$ is smooth map, as continuous. Then $F \circ \varepsilon$ is smooth. Near to the identity, $F = (F \circ \varepsilon) \circ \varepsilon^{-1}$ smooth, precisely on V . Using the fact that L_g is diffeomorphism, F is smooth everywhere. $\square$

Moreover, the exponential map is also a tool to understand the correspondence of subgroup. One can see is if H is Lie subgroup of G then

$$T_e H = \{v \in T_e G : \forall t \in \mathbb{R} : \exp_G(tv) \in H\}. \quad (3.1.1)$$

If $\mathfrak{h}$ is Lie subalgebra of $\mathfrak{g} = T_e G$, exponential map can tells us a subgroup of G which correspond to $\mathfrak{h}$. To see it, let's start with some important facts.

3.1.8 Theorem. If G is a Lie group and $X, Y \in T_e G$, then

- (a) $\exp(tX) \exp(tY) = \exp\left(t(X + Y) + \frac{t^2}{2}[X, Y] + O(t^3)\right);$
- (b) $\exp(-tX) \exp(-tY) \exp(tX) \exp(tY) = \exp(t^2[X, Y] + O(t^3));$
- (c) $\exp(tX) \exp(tY) \exp(-tX) = \exp(tY + t^2[X, Y] + O(t^3)).$

The proof can be found in [Spivak \(1970\)](#), theorem 14 of chapter 10.

3.1.9 Theorem (Baker–Campbell–Hausdorff). Let G be a Lie group with Lie algebra $\mathfrak{g} = T_e G$ and exponential map $\exp: \mathfrak{g} \rightarrow G$. For $X, Y \in \mathfrak{g}$ sufficiently small there exists a unique element $Z \in \mathfrak{g}$ such that

$$\exp(X) \exp(Y) = \exp(Z).$$

Also:

$$Z = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] - \frac{1}{24}[Y, [X, [X, Y]]] + \cdots,$$

and the RHS is a convergent series.

This a classical result that I will not prove.

The naturality of the exponential map is very powerful. One can see also is if φ is a Lie group morphism $\varphi: H \rightarrow G$ is injective, there are small neighbourhoods U of e_H and V of e_G where $\exp_G: \tilde{V} \rightarrow V$ and $\exp_H^{-1}: U \rightarrow \tilde{U}$ are diffeomorphisms. The naturality tells us $(d\varphi)_e$ is injective on $\tilde{U}$, then injective in all $T_e H$. By Proposition 2.1.6, it is an immersion. Consequently, $\varphi(H)$ is Lie subgroup of G .

3.1.10 Theorem. Let G Lie group and $\mathfrak{h}$ is Lie subalgebra of $T_e G$. Then, the subgroup H , generated by $\exp_G(\mathfrak{h})$ is a Lie subgroup of G and its tangent space is exactly $\mathfrak{h}$.

Proof. Let H be the subgroup generated by $\exp_G(\mathfrak{h})$ in G . Let Δ be a small ball centred origin in $\mathfrak{g}$ where BCH formula is valid and $\exp_G: \Delta \rightarrow \exp(\Delta)$ is diffeomorphism. Notice that H is generated by a neighbourhood of e , $H_0 = \exp(\Delta \cap \mathfrak{h})$. Since $H_0^{-1} = H_0$, $H = \bigcup_{k \in \mathbb{N}} H_0^k$. We can put topology

τ_0 on H_0 that makes $\Delta \cap \mathfrak{h}$ homeomorphic to H_0 , via $\exp_G$ (must be the one induced from G). One can notice also $H_0 H_0 \cap \exp(\Delta) = H_0$ (recursively $H \cap \exp(\Delta) = H_0$). H has topology such that τ_0 is induced topology on H_0 and all left translations L_h are homeomorphisms. With this topology, there is a unique smooth structure on H which the translations are also diffeomorphisms and the structure restricted on H_0 is the same the one from G . Noticing that $\mathfrak{h}$ is submanifold of $\mathfrak{g}$, $\Delta \cap \mathfrak{h}$ is submanifold of Δ then for H_0 is submanifold of $\exp(\Delta)$, so submanifold of G . So the differential of the inclusion $H \hookrightarrow G$ is injective at the points of H_0 , then for any point, as the translations are diffeomorphisms. Moreover, clearly $T_e H = T_e H_0 = \mathfrak{h}$. $\square$

Now, one can ask, does any Lie algebra morphism between tangent spaces of Lie groups are coming from Lie group morphism. In another does the functor $\mathbb{T}$ defined at (2.3.3) is full. From Example 2.1.18, $(d\phi)_e$ is an isomorphism from $T_e \mathbf{SU}(2)$ to $\mathbf{SO}(3, \mathbb{R})$. Then $f = (d\phi)_e^{-1}$ is an isomorphism from $\mathbf{SO}(3, \mathbb{R})$ to $\mathbf{SU}(2)$. If its is coming from Lie group morphism $\varphi : \mathbf{SO}(3, \mathbb{R}) \rightarrow \mathbf{SU}(2)$, φ have to be covering map from Theorem 2.1.17. That is contradiction because $\mathbf{SU}(2)$ is simply connected but $\mathbf{SO}(3, \mathbb{R})$ connected but not simply connected. Hence, $\mathbb{T}$ is not full.

3.1.11 Theorem (Lie's Second Theorem). Let G, H two Lie groups, G simply connected and $\phi : T_e G \rightarrow T_e H$ a Lie algebra morphism. Then there is a unique Lie group morphism $\varphi : G \rightarrow H$ such that $(d\varphi)_e = \phi$.

Proof. The graph of ϕ , $\text{graph}(\phi)$ is clearly a subspace of $T_{(e,e)}(G \times H) = T_e G \times T_e H$. It is closed under bracket, as ϕ is Lie algebra morphism, then Lie subalgebra. By Theorem 3.1.10, there is K Lie subgroup of $G \times H$ such that $T_{(e,e)} K = \text{graph}(\phi)$. The differential of the projection $\pi_1 : K \rightarrow G$ at identity can be thought to be the projection $\tilde{\pi}_1$ of $\text{graph}(\phi)$ to $T_e G$, that is an isomorphism. Then π_1 is a covering map, necessarily isomorphism of Lie groups because G is simply connected. Now, define $\varphi = \pi_2 \circ \pi_1^{-1}$ whose differential can be thought $\tilde{\pi}_2 \circ \tilde{\pi}_1^{-1} = \phi$. $\square$

The main ideas of the proofs of Theorems 3.1.10 and 3.1.11 are mostly taken from Fulton and Harris (2013), I have tried to provide details and solve some steps that the author let as an exercises. As comments, from Theorem 3.1.10, the connected component of identity of such H is the unique connected Lie subgroup H' that satisfies $T_e H' = \mathfrak{h}$ because H' has to contains $\exp(\mathfrak{h})$. One another important consequence is the Lie's third theorem. First, let's state the Ado's theorem.

3.1.12 Theorem (Ado's Theorem). Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field $\mathbb{K}$ of characteristic zero (e.g., $\mathbb{R}$ or $\mathbb{C}$). Then there exists a finite-dimensional vector space V over $\mathbb{K}$ and an injective Lie algebra homomorphism

$$\varphi : \mathfrak{g} \hookrightarrow \mathfrak{gl}(V).$$

The proof of this theorem is seen, as appendix in Fulton and Harris (2013) at appendix E.

3.1.13 Theorem (Lie's third theorem). Every finite-dimensional real Lie algebra $\mathfrak{g}$ is the Lie algebra of some simply connected Lie group.

Proof. Ado's theorem tell us that $\mathfrak{g}$ is isomorphic to $\mathfrak{h}$, which is a Lie subalgebra of $\mathfrak{gl}(\mathbb{R}^n) = T_e \mathbf{GL}(n, \mathbb{R})$. By Theorem 3.1.10, there is a connected Lie subgroup H of $\mathbf{GL}(n, \mathbb{R})$ such that $T_e H = \mathfrak{h}$. Then the universal cover of H , is the candidate. $\square$

We get now, the functor $\mathbb{T}$ induces an isomorphism of categories between the category of the simply connected Lie groups $\mathcal{SCLG}$ and category the finite dimensional Lie algebras $\mathcal{FLA}$.

3.2 Some words on classifications

In this part, we talk some initial classification of Lie algebras. Basically, we just state some definition with main propositions with some proofs.

If we ask ourselves when a connected Lie group G is simple (has no non-trivial normal subgroup), it is not always easy to check. But the Lie correspondences allow us to study the Lie algebra associated.

First, let H a connected Lie subgroup of G . If it is normal then for any $v \in T_e H$ and $w \in T_e G$,

$$\gamma(t, s) = \exp_G(sw) \exp_G(tv) \exp_G(-sw) \in H, \gamma(0) = e \implies [u, v] = \left. \frac{d}{dt} \right|_{t=0} \left. \frac{d}{ds} \right|_{s=0} \gamma(s, t) \in T_e H.$$

So $T_e H$ is an ideal of $T_e G$. The converse is also true. Indeed, let $T_e H$ be an ideal of $T_e G$ and $u \in T_e H$ $a \in G$. Considering the conjugation $\psi_a : H \rightarrow G$ and the naturality of the exponential map we get

$$a \exp_H(tX) a^{-1} = \exp_G(Ad(a)x) \in H.$$

So $aHa^{-1} \subset H$ is true in neighbourhood of H , by connectedness true in all H .

Now, then, the natural way to define a simple Lie algebra is a Lie algebra with no non trivial ideals.

One another trivial fact is that if G is abelian then $T_e G$ has bracket zero every where. Moreover, if, $T_e G$ has bracket zero then $\exp_G(u) \exp_G(v) = \exp_G(u + v) = \exp_G(v + u) = \exp_G(v) \exp_G(u)$. Then G commutes on neighbourhood in G . Again, by connectedness, commutes everywhere. So the right definition of abelian Lie algebra is having zero bracket. There are some class of Lie algebra still interesting even though they are not simple neither abelian.

3.2.1 Definition. Let $\mathfrak{g}$ be a Lie algebra. The lower central series $\mathcal{D}_k \mathfrak{g}$ and the derived series $\mathcal{D}^k \mathfrak{g}$ are defined recursively as follow: $\mathcal{D}_0 \mathfrak{g} = \mathcal{D}^0 \mathfrak{g} = \mathfrak{g}$,

$$\mathcal{D}_k \mathfrak{g} = [\mathfrak{g}, \mathcal{D}_{k-1} \mathfrak{g}] \quad \text{and} \quad \mathcal{D}^k \mathfrak{g} = [\mathcal{D}^{k-1} \mathfrak{g}, \mathcal{D}^{k-1} \mathfrak{g}].$$

Clearly $\mathcal{D}^k \mathfrak{g} \subseteq \mathcal{D}_k \mathfrak{g}$. It is checkable also that they are ideals.

3.2.2 Definition (Fulton and Harris (2013)). (a) We say that $\mathfrak{g}$ is *nilpotent* if $\mathcal{D}_k \mathfrak{g} = 0$ for some k .

(b) We say that $\mathfrak{g}$ is *solvable* if $\mathcal{D}^k \mathfrak{g} = 0$ for some k .

(c) We say that $\mathfrak{g}$ is *perfect* if $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

(d) We say that $\mathfrak{g}$ is *semisimple* if $\mathfrak{g}$ has no nonzero solvable ideals.

3.2.3 Theorem (Engel's Theorem). Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be any Lie subalgebra such that every $X \in \mathfrak{g}$ is a nilpotent endomorphism of V . Then there is a nonzero vector $v \in V$ such that $X(v) = 0$ for all $X \in \mathfrak{g}$.

Proof by induction on $n = \dim(\mathfrak{g})$. For $n = 1$, it suffices to take a basis X and the smallest positive integer k satisfying $X^k = 0$ (this k must be greater than 1, otherwise $X = 0$). Then there is $v \in V \setminus \{0\}$ such that $X^{k-1}(v) = 0$. This v does the job.

Now, assume $\dim(\mathfrak{g}) \geq 2$. Let $\mathfrak{h}$ a maximal proper Lie subalgebra of $\mathfrak{g}$. $\text{ad}(H)$ induces an endomorphism of $\mathfrak{g}/\mathfrak{h}$ because $\mathfrak{h}$ is Lie subalgebra of $\mathfrak{g}$. Set $W := \mathfrak{g}/\mathfrak{h}$ and $\mathfrak{k} := \{\overline{\text{ad}}(H) : W \rightarrow W, H \in \mathfrak{h}\}$. Clearly $\mathfrak{k}$ is a linear subspace of $\mathfrak{gl}(W)$ that embeds in $\mathfrak{h}$, then at most of dimension $\dim(\mathfrak{h}) < \dim(\mathfrak{g})$. Notice that, $[\text{ad}(H), \text{ad}(H')](x) = [H, [H', x]] - [H', [H, x]] = [H, H'], x = \text{ad}([H, H'])(x)$, for any $x \in \mathfrak{g}$. It induces the equality in W , then $\mathfrak{k}$ is a Lie subalgebra of $\mathfrak{gl}(W)$. By induction, there is nonzero $\overline{X} \in W$ such that for any $K \in \mathfrak{k}$, $K(\overline{X}) = 0$. That means there is $X \in \mathfrak{g} \setminus \mathfrak{h}$ such that for any $H \in \mathfrak{h}$, $\text{ad}(H)(X) \in \mathfrak{h}$. By maximality of $\mathfrak{h}$, $\langle \mathfrak{h}, X \rangle = \mathfrak{g}$, then $\mathfrak{h}$ is an ideal of codimension 1.

Now, we use induction again to say the linear subspace $U := \{v \in V : \forall H \in \mathfrak{h}, H(v) = 0\} \neq \{0\}$. We have $X[U] \subset U$. Indeed, for all $u \in U$ and $H \in \mathfrak{h}$, $H(X(u)) = X(H(u)) + [X, H](u) = 0$, that implies $X(u) \in U$. As X is nilpotent on V , then also on U . So there is $0 \neq v \in U$ such that $X(v) = 0$. $\square$

This gives a characterization of nilpotent Lie algebra $\mathfrak{g}$: it is necessary and sufficient that $\text{ad}(X) \in \mathfrak{gl}(\mathfrak{g})$ is nilpotent for any $X \in \mathfrak{g}$.

3.2.4 Corollary. A nilpotent Lie algebra $\mathfrak{g}$ is isomorphic to a subalgebra of the nilpotent Lie algebra in $\mathfrak{gl}(m, \mathbb{R})$ of upper triangular matrices with zeros along the diagonals.

It is consequence of Ado's theorem and Engel's theorem.

3.2.5 Theorem (Lie's theorem, [Fulton and Harris \(2013\)](#)). Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ complex solvable Lie algebra. Then there exists a nonzero vector $v \in V$ that is eigenvector of all $X \in \mathfrak{g}$.

The proof is similar to the Engel's theorem. To get an ideal of codimension one, we can consider the quotient map $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$, with abelian codomain. Just take non trivial linear functional $f : \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \rightarrow \mathbb{K}$ and $\mathfrak{h} = \text{Ker}(f \circ \pi)$ is an ideal of codimension one. By induction the linear subspace $U := \{u \in V : \forall H \in \mathfrak{h}, H(u) = \lambda_H u\} \neq \{0\}$. Now, we take $X \in \mathfrak{g} \setminus \mathfrak{h}$ and claim that $X(v) = \lambda u$, for some u to conclude there is v nonzero vector in U that is eigenvector² of $X|_U$.

From this theorem, we can say that for any solvable Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}(V)$, there exists basis of V where $\mathfrak{g}$ is Lie subalgebra of upper triangular matrices.

3.2.6 Proposition. A lie algebra $\mathfrak{g}$ is solvable if and only if the Killing³ form is zero on the commutator $[\mathfrak{g}, \mathfrak{g}]$ if and only $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

3.2.7 Theorem (Cartan's Second Criterion). A Lie algebra $\mathfrak{g}$ is semisimple if and only if the Killing form is nondegenerate.

3.2.8 Theorem (Preservation of Jordan Decomposition, [Fulton and Harris \(2013\)](#)). Let $\mathfrak{g}$ be a semisimple Lie algebra. For any $X \in \mathfrak{g}$, there is X_s and $X_n \in \mathfrak{g}$ such that for any representation $\varphi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, we have

$$(\varphi(X))_s = \varphi(X_s) \text{ and } (\varphi(X))_n = \varphi(X_n).$$

In other words, if we think of ρ as injective and $\mathfrak{g}$ accordingly as a Lie subalgebra of $\mathfrak{gl}(V)$, the diagonalizable and nilpotent parts of any element X of $\mathfrak{g}$ are again in $\mathfrak{g}$ and are independent of the particular representation ρ .

3.3 Relation with exponential map in Riemannian manifolds

One natural question for us is, why it is called exponential map. When it meets with definition in Riemannian manifold Exp_e . In this section we will discuss the importance of having bi-invariant metric that guarantees that the both exponential maps are the same.

Let's note first that if they coincide for some metric $\langle \cdot, \cdot \rangle$ in G then Exp_e is defined everywhere, then by Hopf-Rinow theorem they are surjective if G is connected. So if the exponential map $\exp_G$ is not surjective, we can not expected to find a Riemannian metric on G so that $\text{Exp}_e = \exp_G$.

The reference that inspire me to write this part is [Alexandrino et al. \(2015\)](#).

²Because, we are in complex case.

³See definition [3.3.6](#).

3.3.1 Example. Recall from Example 3.1.6, for linear group the exponential map is the exponential of matrices. Also, using the relation (3.1.1), any element in $\mathfrak{h} = T_e \mathbf{SL}(2, \mathbb{R})$ is of form $A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$.

We can see that there is no $A \in \mathfrak{h}$ such that $e^A = X$, $X = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$. Otherwise, $AX = XA$ then $a = c = 0$. Then $e^A = A$, contradiction. This tell us the exponential map is not surjective. Then, we can not put metric on $\mathbf{SL}(2, \mathbb{R})$ making the exponential maps to coincide.

One can expect they should be the same if the one-parameter subgroups are the geodesics. One thing is, it is verified if the metric is bi-invariant. We mean by a metric to be left-invariant if the left translations are isometry i.e.

$$\forall g, h \in G, u, v \in T_h G : \quad \langle (dL_g)_h(u), (dL_g)_h(v) \rangle_{gh} = \langle u, v \rangle_h. \quad (3.3.1)$$

Similarly for the right-invariant and the bi-invariant metric is metric that is left-invariant and right-invariant. We note that each inner product at $T_e G$, we can construct a left-invariant (or right) metric. Indeed, if $\langle, \rangle$ is inner product of $T_e G$

$$Q_g(u, v) := \langle (dL_{g^{-1}})_g(u), (dL_{g^{-1}})_g(v) \rangle$$

varies smoothly and satisfies (3.3.1) a left-invariant metric. But we can not always have bi-invariant metric on an arbitrary Lie group G .

3.3.2 Proposition. Let $\langle, \rangle$ be bi-invariant Riemannian metric on G . Then we have

$$\text{Exp}_e = \exp_G.$$

Proof. Let ∇ the Levi-Civita connection of $\langle, \rangle$ and $u, v, w \in T_e g$ with left-invariant vector fields respective X, Y, Z . The compatibility of ∇ gives us

$$\begin{aligned} \langle \nabla_Y X, Z \rangle &= \frac{1}{2} (X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle [X, Z], Y \rangle - \langle [Y, Z], X \rangle - \langle [X, Y], Z \rangle) \\ &= \frac{1}{2} (-\langle [X, Z], Y \rangle - \langle [Y, Z], X \rangle - \langle [X, Y], Z \rangle). \end{aligned} \quad (3.3.2)$$

Also $\psi_g = L_g R_{g^{-1}}$, then $\text{Ad}(g) = (dL_g)_{g^{-1}}(dR_{g^{-1}})_e$. So

$$\begin{aligned} \langle \text{Ad}(\exp(tu))v, \text{Ad}(\exp(tu))w \rangle &= \langle v, w \rangle && \text{as the metric is bi-invariant,} \\ \implies \langle [u, v], w \rangle + \langle v, [u, w] \rangle &= 0 && \text{after differentiating.} \\ \implies \nabla_Y X = \frac{1}{2}[Y, X] \text{ and } \nabla_X X = 0 &&& \text{by (3.3.2).} \end{aligned} \quad (3.3.3)$$

Now if γ_u is the one-parameter subgroup relative to $u \in T_e G$ and X is the left-invariant vector field relative to u . We have $X(\gamma'_u(t)) = \gamma_u(t)$, then $\frac{D}{dt} \gamma'_u(t) = \nabla_X X = 0$. So γ_u is geodesic. The uniqueness ensures us that geodesics are the one-parameter subgroups (we shall see that if γ is geodesic, it is an integral curve of the left-invariant vector field associated to $\gamma'(0)$). $\square$

Then the class of Lie groups that admit bi-invariant metric, is one of important of Lie group. Let's start to see one condition that ensures us the existence of such metric.

3.3.3 Proposition. Every compact Lie group admits a bi-invariant metric q .

Before starting the proof, a n -form⁴ ω is right-invariant if $(R_h)^*\omega = \omega$ for all $h \in G$ i.e.

$$\forall g, h \in G, u \in (T_g G)^n : \quad \omega_{gh}((dR_h)_g(u_1), \dots, (dR_h)_g(u_n)) = \omega_g(u_1, \dots, u_n). \quad (3.3.4)$$

One can show is, we can always construct such n -form, as they are clearly uniquely determined by its value at the identity. We can take any alternating multilinear $\omega_e : (T_e G)^n \rightarrow \mathbb{R}$ and define

$$\omega_g(u_1, \dots, u_n) := \omega_e((dR_{g^{-1}})_g(u_1), \dots, (dR_{g^{-1}})_g(u_n)).$$

It varies smoothly on g and satisfies (3.3.4).

Proof of Proposition 3.3.3. Consider a non-vanishing n -form ω and $\langle, \rangle$ a right invariant metric on G . One can define, for any $u, v \in T_a G$

$$q_a(u, v) := \int_G f_a(g) \omega, \quad \text{for } f_a(g) := \langle (dL_g)_a(u), (dL_g)_a(v) \rangle_{ga}. \quad (3.3.5)$$

As G is compact, this integral is finite. Also, clearly it is inner product on $T_a G$ and varies smoothly on G . Then it is Riemannian metric and we claim that it is bi-invariant. Remarking that the left translation and right translation commute, hence their differentials, we get quickly that the metric is right invariant. To see that is left-invariant, using $L_{gh} = L_g L_h$,

$$\begin{aligned} q_{ha}((dL_h)_a(u), (dL_h)_a(v)) &= \int_G \langle L_{gh}u, L_{gh}v \rangle_{gha} = \int_G f(g) \omega \\ &= \int_G (f_a \circ R_h)(g) (R_h)^* \omega && \text{as } \omega \text{ is right invariant,} \\ &= \int_G R_h^*(f_a \omega) = \int_G f_a(g) \omega && \text{as } R_h \text{ is diffeomorphism and } G \text{ orientable.} \end{aligned}$$

Hence q is bi-invariant metric. □

First, we notice that if the right-invariant $\langle, \rangle$ chosen in the proof above is already bi-invariant, this construction gives the same metric up to constant.

Now, if using (3.3.3), the tensor curvature is computable by

$$R(u, v)w = \frac{1}{4} [[u, v], w] \implies R(u, v, u, v) = \frac{1}{4} ||[u, v]||^2. \quad (3.3.6)$$

Hence, sectional curvature respect to bi-invariant metric is always non-negative at identity, hence everywhere by homogeneity (L_g and R_g are isometries).

3.3.4 Example (Sectional curvature on $\mathbf{SO}(3, \mathbb{R})$). First, from (2.1.5), $M_k = -\frac{1}{2}N_k$, the isomorphism

$$\begin{aligned} \mathbb{B} : \mathbb{R}^3 &\longrightarrow T_e \mathbf{SO}(3, \mathbb{R}) = \text{span}_{\mathbb{R}}\{M_1, M_2, M_3\} \\ (x_1, x_2, x_3) &\longmapsto \mathbb{B}((x_1, x_2, x_3)) = x_1 M_1 + x_2 M_2 + x_3 M_3, \end{aligned}$$

respects the bracket if we consider the cross product on $\mathbb{R}^3$. Let's try to take a nice inner product on $T_e G$. We have the trivial on $\mathbb{R}^3$, we can push-forward it on $T_e G$

$$\langle x, y \rangle = u \cdot y \implies \langle u, v \rangle_e = \langle \mathbb{B}(x), \mathbb{B}(y) \rangle := \sum_k x_k y_k = -\frac{1}{2} \text{Tr}(uv).$$

⁴ n -form $\omega = (\omega_x : (T_x G)^n \rightarrow \mathbb{K})_{x \in G}$, where ω_x is a multilinear, alternating and $x \mapsto \omega_x$ are smooth (holomorphic for holomorphic n -form).

Then the right-invariant metric associated is

$$\langle u, v \rangle_g = \langle (dR_{g^{-1}})_g(u), (dR_{g^{-1}})_g(v) \rangle_e = -\frac{1}{2} \text{Tr}(ug^{-1}vg^{-1}). \quad (3.3.7)$$

Following the things learned so far, one can try to find also right-invariant 3-form. Again, we can choose a nice alternating trilinear on $\mathbb{R}^3$, that is the determinant and push-forward it on $T_e G$ and get

$$\omega_e(u, v, w) = \omega_e(\mathbb{B}(x), \mathbb{B}(y), \mathbb{B}(z)) = \det(x, y, z) = x \cdot (y \wedge z) = -\frac{1}{2} \text{Tr}(u, [v, w]).$$

We should apply (3.3.5) to get bi-invariant metric, but one check $\langle \cdot, \cdot \rangle_g$ defined at (3.3.7) is already bi-invariant. So with this metric, the exponential maps coincide (at identity). Moreover, by construction, M_1, M_2, M_3 form an orthogonal basis of $T_e \mathbf{SO}(3, \mathbb{R})$ and $[M_k, M_l] = \mathbb{B}(e_k \wedge e_l) = \mathbb{B}(\varepsilon_{klm} e_m) = \varepsilon_{klm} M_m$. Applying (3.3.6), $R_{jklm} = R(M_j, M_k, M_l, M_m) = \frac{1}{4} \langle [\varepsilon_{jks} M_s, M_l], M_m \rangle_e$. Then

$$R_{jklm} = \frac{1}{4} \sum_{s=1}^3 \varepsilon_{jks} \varepsilon_{slm} \implies R_{1212} = R_{1313} = R_{2323} = \frac{1}{4}.$$

So the sectional curvature is $\frac{1}{4}$.

But also metric that is right-invariant and Ad-invariant must be bi-invariant. In particular, for abelian group, $\psi_g = Id_G$ and $\text{Ad}(g) = Id_{\mathfrak{g}}$, then we can always find a bi-invariant. In another word, $L_g = R_g$, then being right-invariant is equivalent to be left-invariant.

Now, we have two class of Lie groups that admit bi-invariant metric. It is straightforward to see that the product metric of bi-invariant metrics is bi-invariant.

One can ask is if they are all.

3.3.5 Theorem. A connected Lie group G admits a bi-invariant metric if and only if it is isomorphic to the cartesian product of a compact group and a vector space.

The proof of this theorem is seen in in [Milnor \(1976\)](#).

Trying to study the existence of bi-invariant metric and to see if the Lie algebra $T_e G$ is semisimple, there is tools that we need to talk: the Killing form.

3.3.6 Definition (Killing form). Let G Lie group. The Killing form of $\mathfrak{g} = T_e G$ (also said Killing form of G) is the symmetric bilinear form

$$B(u, v) := \text{Tr}(\text{ad}(u) \text{ad}(v)).$$

Since $\text{Ad}(g^{-1})$ is an automorphism of Lie algebra, $\text{ad}(\text{Ad}(g^{-1})(u)) \text{Ad}(g^{-1})(v) = \text{Ad}(g^{-1})(\text{ad}(u)(v))$. Then $\text{Ad}(g) \circ \text{ad}(\text{Ad}(g^{-1})(u)) \circ \text{Ad}(g^{-1}) = \text{ad}(u)$. So $\text{ad}(\text{Ad}(g)(w)) = \text{Ad}(g) \circ \text{ad}(w) \circ \text{Ad}(g)^{-1}$. This fact implies that the killing form is Ad-invariant.

Consequently, if G is semisimple with negative-definite Killing form B , it admits bi-invariant metric $-B$. One can see also is now, if G admit a bi-invariant metric Q , the Ricci tensor does not depend on Q . Indeed, $\text{Ric}(u, v) = \text{Tr}((R(u, -)v)) = \text{Tr}(Tr)(\frac{1}{4}[[u, -], v]) = -\frac{1}{4} \text{Tr}(Tr)([v, [u, -]]) = -\frac{1}{4} B(u, v)$.

3.3.7 Proposition. For a semisimple connected Lie group G , being compact is equivalent to having Killing form negative-definite B .

Proof. See [Alexandrino et al. \(2015\)](#), Theorem 2.35. □

3.3.8 Proposition ([Alexandrino et al. \(2015\)](#)). Let G be a compact simple Lie group. Then the bi-invariant metric is unique up to rescaling.

3.4 Lie Groups with Left-Invariant Metrics and Cartan Connections

In this section, we will equip G is a left-invariant metric. Recall that the left-invariant metric is uniquely determined by an inner product at identity. Then, identifying $T_e G$ as the left-invariant vector field $\mathcal{L}(G)$, we have

$$\langle X(g), Y(g) \rangle_g = \langle X(e), Y(e) \rangle_e.$$

So it is enough to study the inner product at the Lie algebra $\mathfrak{g} = T_e G$.

Recall that the equation (3.3.2) is still available for any Riemannian connection. We may need to choose an orthonormal basis $e_1, \dots, e_n$ and some notation: $\alpha_{ijk} := \langle [e_i, e_j], e_k \rangle$ so that

$$\nabla_{e_i} e_j = \frac{1}{2} \sum_k (\alpha_{ijk} - \alpha_{jki} + \alpha_{kij}) e_k.$$

From this one, on can compute the sectional curvature

$$\kappa(e_1, e_2) = \sum_k \left(\frac{1}{2} \alpha_{12k} (-\alpha_{12k} + \alpha_{2k1} + \alpha_{k12}) - \frac{1}{4} (\alpha_{12k} - \alpha_{2k1} + \alpha_{k12}) (\alpha_{12k} + \alpha_{2k1} - \alpha_{k12}) - \alpha_{k11} \alpha_{k22} \right).$$

From this equality, we can deduce that if for any $\text{ad}(e_1)$ is skew-adjoint, then for any $k \neq 1$, the $\kappa(e_1, e_k) = \sum_j \frac{1}{4} (\alpha_{kj1})^2$. This can be generalise if $\text{ad}(u)$ is skew-symmetric then for all v , $\kappa(u, v) \geq 0$.

3.4.1 Lemma. Let $\mathfrak{u}$ be an ideal of dimension $n - 1$ and $x \in \mathfrak{g} \setminus \mathfrak{u}$. Let L be an endomorphism of $\mathfrak{u}$ defined by $L(u) := [x, u]$ whose its adjoint transformation is denoted by L^* ⁵. Denoting $S = \frac{1}{2}(L + L^*)$ the self adjoint part of L and $\bar{\nabla}$, the Riemannian connection for the metric(inner product), restricted on $\mathfrak{u}$, for any $u, v \in \mathfrak{u}$

$$\nabla_x u = \frac{1}{2}(L - L^*)u, \quad \nabla_x x = 0, \quad \nabla_u x = -Su \text{ and } \nabla_u v = \bar{\nabla}_u v + \langle Su, v \rangle x. \quad (3.4.1)$$

The proof is mostly from the equation (3.3.2).

3.4.2 Proposition. Let x be a unit vector of $\mathfrak{g} = T_e G$, orthogonal to the commutator $[g, g]$. Then the Ricci curvature $r(x)$ is non-positive. It is zero if and only if $\text{ad}(x)$ is skew-adjoint.

Proof. Let $\mathfrak{u}$ the subspace consisting of $y \in \mathfrak{g}$ such that $\langle x, y \rangle = 0$. Clearly $[\mathfrak{g}, \mathfrak{g}] \subset \mathfrak{u}$. Then $\mathfrak{u}$ is an ideal of dimension $n - 1$, so we can apply Lemma 3.4.1. Now we consider the basis $y_1, \dots, y_{n-1}$ the $n - 1$ unit vectors linearly independent, eigenvector of S , namely $S(y_i) = \lambda_i y_i$, for each $i = 1, \dots, n - 1$. As $\lambda_i = \langle S(y_i), y_i \rangle = \frac{1}{2} (\langle L(y_i), y_i \rangle + \langle L^*(y_i), y_i \rangle)$, $\lambda_i = \langle L(y_i), y_i \rangle = \langle y_i, L^*(y_i) \rangle$. We use this fact with (3.4.1) and get the sectional curvatures $k(x, y_i) = -\lambda_i^2$. Hence $r(x) = -\text{Tr}(S^2)$. Moreover, the trace of the square of a self-adjoint operator S is zero if and only if $S = 0$. Therefore, $r(x) = 0$ if and only if L is skew-adjoint. $\square$

A left Haar measure on G is a radon measure μ on the Borel σ -algebra $\mathcal{B}(G)$ ⁶ such that $\mu(gS) = \mu(S)$ for all $g \in G$ and $S \in \mathcal{B}(G)$. It exists always because there is always a non trivial element in $\wedge^n(T_e G)$ (we discuss it at 3.3.3) that induces a n -form left-invariant ω and we can characterize by

$$\mu(S) = \int_S \omega.$$

It is radon measure, left-invariant and it is this the only way to get it, coming from an element of $\wedge^n(T_e G)$. Then as $\wedge^n(T_e G)$ is of dimension one, then the left Haar measure is unique up to scaling.

⁵Recall that is the unique endomorphism satisfying $\langle Lu, w \rangle = \langle u, L^*w \rangle$ for any $u, w \in T_e G$.

⁶ σ -algebra generated by the open sets.

3.4.3 Definition. A Lie group G is unimodular if the left Haar measure μ is right-invariant, namely $\mu(Sg) = \mu(S)$ for all $g \in G$ and $S \in \mathcal{B}(G)$. A Lie algebra $\mathfrak{h}$ is unimodular if for any $h \in \mathfrak{h}$, $\text{Tr}((\text{ad}(h))) = 0$.

It is proven in Helgason (2024) (page 366) that G is unimodular if and only if the adjoint representation $\text{Ad}(g)$ has a determinant ± 1 for any $g \in G$. It follows that semisimple Lie groups are unimodular, because $(\text{Ad}(g))^2 = 1$, as the nondegenerate Killing form B is Ad -invariant. Another quick consequence of this fact is, the compact Lie groups are unimodular. Indeed, the set consisting of the $|\text{Ad}(g)|$, can be view as image of continuous group morphism from G to the positive real numbers. Then it is compact subgroup of $\mathbb{R}_+^*$, so have to be $\{1\}$. The naturality of exponential map ensures us that for connected Lie group, G is unimodular if and only if $T_e G$ is unimodular.

3.4.4 Proposition. If all Ricci curvatures in a connected Lie group G are non-negatives, then G is unimodular.

Proof. Let $x \in \mathfrak{g} = T_e G$. Then it is impossible to have $r(x) < 0$. By Proposition 3.4.2, x is not orthogonal to $[g, g]$ or $\text{ad}(x)$ is skew adjoint. But Jacobi identity gives us $\text{ad}([g_1, g_2]) = \text{ad}(g_1)\text{ad}(g_2) - \text{ad}(g_2)\text{ad}(g_1)$ that implies any element of $[\mathfrak{g}, \mathfrak{g}]$ in the unimodular kernel $\mathfrak{u} = \{u \in \mathfrak{g} : \text{Tr}(\text{ad}(u)) = 0\}$. Then x can not be orthogonal to $\mathfrak{u}$ and $\text{Tr}(\text{ad}(x)) = 0$. So $x \in \mathfrak{u}$. Hence $\mathfrak{g} = \mathfrak{u}$. $\square$

3.4.5 Theorem (Milnor (1976)). If a connected Lie group G has a left-invariant metric with all sectional curvatures $\kappa \leq 0$, then it is solvable. If G is unimodular, then any such metric with $K \leq 0$ must actually be flat ($\kappa \equiv 0$).

3.4.6 Example. Note that the semi-product of Lie groups has Lie group structure. In particular, $G = \mathbb{R} \rtimes \mathbb{R}_+^*$ is Lie group with the law $(x_1, y_1) \cdot (x_2, y_2) = (x_1 + y_1 x_2, y_1 y_2)$. The neutral element is $e = (0, 1)$ and inverse of (x, y) is $(-\frac{x}{y}, \frac{1}{y})$. Consider the usual inner product $\langle -, - \rangle$ on $T_e G = \mathbb{R}^2$ and let Q the left-invariant metric correspondent. We can check also that that the differential of the left translation is $(dL_{(a,b)})_{(x,y)} = bI_2$. Then

$$Q_{(a,b)}(u, v) = \langle (dL_{(-\frac{a}{b}, \frac{1}{b})})_{(a,b)}(u), (dL_{(-\frac{a}{b}, \frac{1}{b})})_{(a,b)}(v) \rangle = \frac{u \cdot v}{b^2}.$$

It is the hyperbolic $\mathbb{H}^2$ metric, so the sectional curvature is $\kappa = -1$. Hence G is solvable.

Another way to see that is to identify it by set of matrix of form $\begin{pmatrix} y & x \\ 0 & 1 \end{pmatrix}$, $y \in \mathbb{R}$ and $x \in \mathbb{R}_+^*$.

Viewing as $\mathbb{H}^n$ allows us to conclude also that it is not unimodular.

3.4.7 For three-dimensional case

We assume that G is of dimension 3. Let $\langle -, - \rangle$ be an inner product on a vector space $T_g G$.

For $u, v \in T_g G$, there is a vector $w \in T_g G$ such that

$$\langle u, w \rangle = \langle v, w \rangle = 0 \tag{3.4.2}$$

$$|w|^2 = |u|^2 |v|^2 - \langle u, v \rangle^2. \tag{3.4.3}$$

First, if u, v are linearly dependant, (3.4.3) forces w to be zero, that respects also (3.4.2). If they are not the set of all vector z verifying (3.4.2) is subspace of $T_g G$, of codimension 2, then dimension one. With the condition (3.4.3), we have two choice $\pm w$. Now, if we fix orientation on $T_g G$, have only one choice that makes (u, v, w) oriented. such w is the general definition of cross product $u \times v$.

3.4.8 Lemma. Let G Lie group. There is a linear endomorphism L of $T_e G$ such that

$$[u, v] = L(u \times v). \quad (3.4.4)$$

Moreover, L is self adjoint if and only if G is unimodular.

It is because we can choose an oriented orthonormal basis b_1, b_2, b_3 and set an endomorphism of $\mathfrak{g}$,

$$L(b_i \times b_j) = [e_i, b_j] = \text{ad}(b_i)(b_j) = -\text{ad}(b_j)(b_i). \quad (3.4.5)$$

It is defined on basis and it extends uniquely in all $\mathfrak{g}$ satisfying (3.4.4). If $L = (l_{ij})_{i,j}$, (3.4.5) says also $\text{ad}(b_1)(x_1 b_1 + x_2 b_2 + x_3 b_3) = (l_{31} - l_{21})b_1 + (l_{32} - l_{22})b_2 + (l_{33} - l_{23})b_3$, then $\text{Tr}(\text{ad}(e_1)) = -l_{23} + l_{32}$. Doing the same for b_2, b_3 we get $\text{Tr}(\text{ad}(b_2)) = l_{13} - l_{31}$ and $\text{Tr}(\text{ad}(b_3)) = -l_{12} + l_{21}$. Hence L is self adjoint⁷ if and only if $\text{Tr}(\text{ad}(x)) = 0$ for any $v \in \mathfrak{g}$ if and only if $\mathfrak{g}$ is unimodular.

Now, let's focus for the case where G is unimodular (L is self adjoint) and choose an orthonormal basis e_1, e_2, e_3 consisting of eigenvectors of L , with eigenvalues respective λ_i . Then, the, we have

$$[e_2, e_3] = \lambda_1 e_1, \quad [e_3, e_1] = \lambda_2 e_2 \quad \text{and} \quad [e_1, e_2] = \lambda_3 e_3. \quad (3.4.6)$$

Following the approach of Milnor, we denote

$$\mu_i = \frac{1}{2}(\lambda_1 + \lambda_2 + \lambda_3) - \lambda_i.$$

3.4.9 Theorem (Milnor (1976)). The orthonormal basis e_1, e_2, e_3 , chosen as above, diagonalizes the Ricci quadratic form, the principal Ricci curvatures being by

$$r(e_1) = 2\mu_2\mu_3, \quad r(e_2) = 2\mu_1\mu_3 \quad \text{and} \quad r(e_3) = 2\mu_1\mu_2. \quad (3.4.7)$$

Being the trace of the Ricci tensor, the scalar curvature is

$$\rho = 2(\mu_2\mu_3 + \mu_1\mu_3 + \mu_1\mu_2). \quad (3.4.8)$$

Milnor (1976) has many nice examples but I just try to another one. Note that $\text{SO}(2, 1)$ is simple Lie group. Let G is component connected of identity that is then unimodular. Consider the basis $\{e_1, e_2, e_3\}$ of $T_e g = \mathfrak{so}(2, 1)$ be given by:

$$e_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Their Lie brackets are: $[e_1, e_2] = -e_3$, $[e_2, e_3] = e_1$, $[e_3, e_1] = e_2$. We just need to have inner product such that e_1, e_2 and e_3 are orthonormal. Now we consider the left-invariant metric Q corresponding to that inner product, which is $\langle e_i, e_j \rangle = \delta_{ij}$.

Now, to compute the Ricci curvatures and scalar curvature, we just need to compute μ_i .

$$\begin{aligned} \mu_1 &= -\frac{1}{2}, & \mu_2 &= -\frac{1}{2}, & \mu_3 &= \frac{3}{2}, \\ r(e_1) &= 2\mu_2\mu_3 = -\frac{3}{2}, & r(e_2) &= 2\mu_1\mu_3 = -\frac{3}{2}, & r(e_3) &= 2\mu_1\mu_2 = \frac{1}{2}, \\ \rho &= 2(\mu_2\mu_3 + \mu_1\mu_3 + \mu_1\mu_2) = -\frac{5}{2}. \end{aligned}$$

⁷We are in real case.

3.4.10 Theorem (Milnor (1976)). If the connected 3-dimensional Lie group G is not unimodular, then its Lie algebra has a basis e_1, e_2, e_3 so that

$$[e_1, e_2] = \alpha e_2 + \beta e_3, \quad [e_1, e_3] = \gamma e_2 + \delta e_3$$

with $[e_2, e_3] = 0$, and so that the matrix

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

has trace $\alpha + \delta = 2$. If we exclude the exceptional case where A is the identity matrix, then the determinant $D = \alpha\delta - \beta\gamma$ provides a complete isomorphism invariant for this Lie algebra.

Milnor connects this fact to show, for non-unimodular cases, at least the principal Ricci curvatures are negatives.

3.4.11 On Cartan connection

The think that makes the bi-invariant metrics nice, is the fact that $t \mapsto \exp(tu)$ are geodesics. For the left-invariant one we lost this property. Here, we relax that the connection is allowed to be may not compatible, and build another connection that makes the $t \mapsto \exp(tu)$ still geodesics.

3.4.12 Definition. A connection ∇ is left-invariant if

$$\nabla_{(L_a)_*X}((L_a)_*Y) = (L_a)_*(\nabla_X Y) \quad \text{for all } a \in G, X, Y \in \mathfrak{X}(G).$$

Such connection is uniquely determined is value at identity, namely for a bilinear forms $\nabla^{(e)}$ of $T_e G$.

$$(\nabla_X Y)(g) = dL_g(\nabla^{(e)}(X_e, Y_e)).$$

3.4.13 Proposition (Definition). Let ∇ be a left-invariant connection. Then the following assertions are equivalent.

- (a) $\nabla^{(e)}$ is skew-symmetric.
- (b) For any $u \in T_e G$, $t \mapsto \exp(tu)$ are geodesics.

Such connection is called Cartan connection.

Proof. We recall that $t \mapsto \exp(tu)$ is γ_u , the unique smooth curve such that

$$\gamma_u(0) = e, \quad \gamma'_u(0) = u, \quad \text{and } X(\gamma_u(t)) = \gamma'_u(t),$$

where X is left-invariant vector field corresponding to u . Then $\nabla_{\gamma'_u(t)} \gamma'_u(t) = (\nabla_X X)(\gamma_u(t)) = \nabla^{(e)}(u, u)$. This equality ensures the equivalence. $\square$

The first example is $(u, v) \mapsto \lambda[u, v]$ is skew-symmetric then induces a Cartan connection with torsion $T(u, v) = (2\lambda - 1)[u, v]$ and curvature $R(u, v)w = \lambda(1 - \lambda)[[u, v], w]$.

3.4.14 Proposition. For any Lie group G there exists unique symmetric Cartan connection.

Proof. We require to be symmetric then at identity $\nabla^{(e)}(u, v) - \nabla^{(e)}(v, u) = [u, v]$. As $\nabla^{(e)}$ is skew-symmetric, it is equivalent $\nabla^{(e)}(u, v) = \frac{1}{2}[u, v]$. This extends uniquely giving us what we need. $\square$

This connection still allow us to have (3.3.3) and $R(u, v)w = \frac{1}{4}[[u, v], w]$.

Clearly (3.3.3) tells us that if the metric is bi-invariant the unique symmetric connection is the Levi-Civita connection.

4. Representations of Semisimple Lie Algebras

The goal of this part is to present the general approaches to understanding the representations of semisimple complex Lie algebras. We begin with a particular case, the simple Lie algebra $\mathfrak{sl}(n, \mathbb{C})$, and provide examples. There are many ways to see that $\mathfrak{sl}(n, \mathbb{C})$ and $\mathfrak{sp}(n, \mathbb{C})$ are semisimple.

The long-term goal of this part is not only to study complex simple Lie algebras but also Lie groups and real Lie algebras. Here, we start with the complex case, which we complete in this project. We hope that understanding this case first will make it easier to study similar classes later.

4.1 Representations of $\mathfrak{sl}(2, \mathbb{C})$

The structure of $\mathfrak{sl}(2, \mathbb{C})$ is entirely given by

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$[H, X] = 2X, \quad [H, Y] = -2Y, \quad [X, Y] = H.$$

We will learn the representation of $\mathfrak{sl}(2, \mathbb{C})$ by considering the irreducible ones. Also, the focus is finite dimension. So in the next, the representations are finite and irreducible unless stated otherwise.

Let (φ, V) a representation of $\mathfrak{sl}(2, \mathbb{C})$. By Theorem 3.2.8, $\varphi(H)$ is diagonalizable. Then

$$V = \bigoplus_{\alpha \in \text{spec}(H)} V_{\alpha}.$$

For $v \in V_{\alpha}$, one can show is $\varphi(H)(\varphi(X)(v)) = (\alpha+2)\varphi(X)(v)$ and $\varphi(H)(\varphi(Y)(v)) = (\alpha-2)\varphi(Y)(v)$. Then $\varphi(X)$ maps V_{α} to $V_{\alpha+2}$ and $\varphi(Y)$ maps V_{α} to $V_{\alpha-2}$. Irreducibility implies every elements in $\text{spec}(H)$ are modulo 2. By finiteness,

$$V = \bigoplus_{k=0}^{k=N} V_{\alpha_0+2k}. \quad (4.1.1)$$

It is a little bit earlier to be convinced that every step decayed 2 have to be eigenvalue successively but Proposition 4.1.2 makes us sure that if α_0 and $\alpha_0 + 2N$ are eigenvalues then everything elements in between are also eigenvalues.

4.1.1 Lemma. For any $v \in V_{\alpha_0}$,

$$\forall k \in \mathbb{N}^* : \quad \varphi(Y)(\varphi(X)^k(v)) = -k(\alpha_0 + k - 1)\varphi(X)^{k-1}(v). \quad (4.1.2)$$

Proof by Induction. For $k = 1$, considering $\varphi(Y)(v) = 0$, have

$$-\alpha_0 v = [\varphi(Y), \varphi(X)](v) = \varphi(Y)\varphi(X)(v).$$

Assume it is true until k ,

$$\begin{aligned} \varphi(Y)(\varphi(X)^{k+1}(v)) &= [\varphi(Y), \varphi(X)](\varphi(X)^k(v)) + \varphi(X)\left(\varphi(Y)(\varphi(X)^k(v))\right) \\ &= (-\alpha_0 - 2k)\varphi(X)^k(v) - k(\alpha_0 + k - 1)\varphi(X)^k(v) \\ &= -(k+1)(\alpha_0 + k)\varphi(X)^k(v). \end{aligned}$$

Hence (4.1.2). □

4.1.2 Proposition. Let $v \in V_{\alpha_0}$. If $v \neq 0$, $W := \text{span} \left(\{\varphi(X)^k(v) : k \in \mathbb{N}\} \right) = V$.

Proof. Clearly, it is invariant under the action X . By noticing $\varphi(X)^k$ maps V_α to $V_{\alpha+2k}$, it is also invariant under the action of H . Lemma 4.1.1 shows also that it is invariant under the action of Y . Hence W is invariant by φ and by irreducibility $W = V$. $\square$

Now, let M the biggest $m \in \mathbb{N}$ such that $\varphi(X)^m(v) \neq 0$. Then $\varphi(X)^{m+1}(v) = 0$. Applying (4.1.2),

$$0 = \varphi(Y)(\varphi(X)^{m+1}(v)) = -(k+1)(\alpha_0 + m)\varphi(X)^m(v) \implies \alpha_0 = -m \in \mathbb{Z}.$$

Since for $l \neq k$, $\varphi(X)^k(v)$ and $\varphi(X)^l(v)$ lie in different eigenspaces, then $(\varphi(X)^j)_{j=0}^{j=m}$ are linearly independent, so basis. One can deduce also now, if $d = \dim(V)$, $m = d - 1 = N$ and

$$V = V_{1-d} \oplus V_{3-d} \oplus \cdots \oplus V_{d-3} \oplus V_{d-1} =: V^{(d-1)}. \quad (4.1.3)$$

Hence all representations of $\mathfrak{sl}(2, \mathbb{C})$ are of the form (4.1.3).

4.1.3 Example. Let e_1 and e_2 the standard basis of $\mathbb{C}^2$. $\mathbb{C}^2 = \langle e_1 \rangle \oplus \langle e_2 \rangle$. The representation

$$\varphi \left(\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx - ay \end{pmatrix},$$

of $\mathfrak{sl}(2, \mathbb{C})$ on $\mathbb{C}^2$ is checkable irreducible. Then it has the form $V^{(1)} = V_{-1} \oplus V_1$ where $V_1 = \langle e_1 \rangle$ and $V_{-1} = \langle e_2 \rangle$ because $\varphi(H)(e_1) = e_1$ and $\varphi(X)(e_2) = -e_2$.

More generally, the set of homogenous polynomial in $\mathbb{C}[x, y]$ of degree $d - 1$, identified to $\text{Sym}^{d-1}(\mathbb{C}^2)$ is vector space of dimension d , span by $(f_k := x^k y^{n-k})_{k=0}^{k=d-1}$. The endomorphisms of

$$\mathcal{H} = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}; \quad \mathcal{X} = x \frac{\partial}{\partial y} \quad \text{and} \quad \mathcal{Y} = y \frac{\partial}{\partial x}$$

verify $[\mathcal{H}, \mathcal{X}] = 2\mathcal{X}$, $[\mathcal{H}, \mathcal{Y}] = -2\mathcal{Y}$ and $[\mathcal{X}, \mathcal{Y}] = \mathcal{H}$. The linear map $X \mapsto \mathcal{X}$, $Y \mapsto \mathcal{Y}$ and $H \mapsto \mathcal{H}$ is a representation of $\mathfrak{sl}(2, \mathbb{C})$. Also f_k are eigenvectors of $\mathcal{H}$, $\mathcal{H}$ is diagonalisable with $\text{spec}(H) = (2k - n)_{k=0}^{k=d-1}$. One can see also $\mathcal{X}$ maps V_{2k-n} to V_{2k-n+2} and $\mathcal{Y}$ maps V_{2k-n} to V_{2k-n-2} .

Hence this representation is irreducible. Indeed if W is invariant, the restriction of a $\mathcal{H}$ on W is diagonalizable, then contains f_{k_0} for some $k_0 = 0, 1, \dots, d - 1$. Acting with $\mathcal{X}$ and $\mathcal{Y}$, we get $W = \text{Sym}^{d-1}(\mathbb{C}^2)$.

4.1.4 Theorem (Hall (2013)). If (φ, V) is a finite-dimensional representation of $\mathfrak{sl}(2; \mathbb{C})$, not necessarily irreducible, the following results hold.

- (a) Every eigenvalue of $\varphi(H)$ is an integer. Furthermore, if v is an eigenvector for $\varphi(H)$ with eigenvalue λ and $\varphi(X)v = 0$, then λ is a non-negative integer.
- (b) The operators $\varphi(X)$ and $\varphi(Y)$ are nilpotent.
- (c) If we define $S : V \rightarrow V$ by

$$S = e^{\varphi(X)} e^{-\varphi(Y)} e^{\varphi(X)},$$

then S satisfies

$$S\varphi(H)S^{-1} = -\varphi(H).$$

- (d) If an integer k is an eigenvalue for $\varphi(H)$, so is each of the numbers

$$-|k|, -|k| + 2, \dots, |k| - 2, |k|.$$

4.2 Representations of $\mathfrak{sl}(3, \mathbb{C})$

The structure of $\mathfrak{sl}(3, \mathbb{C})$ is entirely given by

$$\begin{aligned} H_1 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & H_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & X_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, & X_3 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ Y_1 &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & Y_2 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & Y_3 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \end{aligned}$$

$$\begin{aligned} [H_1, H_2] &= 0, & [H_i, X_i] &= 2X_i, & [H_2, X_1] &= -X_1, \\ [H_1, X_2] &= -X_2, & [H_1, X_3] &= X_3, & [Y_1, X_3] &= X_2, \\ [H_2, X_3] &= X_3, & [H_i, Y_j] &= -[H_i, X_j], & [Y_2, X_3] &= -X_1, \\ [X_1, X_2] &= X_3, & [Y_2, Y_1] &= Y_3, & [X_3, Y_3] &= H_1 + H_2, \\ [X_i, Y_i] &= H_i, & [X_i, X_3] &= 0, & [Y_i, Y_3] &= 0. \end{aligned}$$

The same for $\mathfrak{sl}(2, \mathbb{C})$, we study an irreducible and finite representation (φ, V) of $\mathfrak{sl}(3, \mathbb{C})$.

One can notice is $\mathfrak{sl}(3, \mathbb{C})$ has the subalgebras $\mathfrak{g}_1 = \langle H_1, X_1, Y_1 \rangle$ and $\mathfrak{g}_2 = \langle H_2, X_2, Y_2 \rangle$, which both isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. The idea is here, try to use the same method like in 4.1, in appropriate way.

As did with $\mathfrak{sl}(2, \mathbb{C})$, we want to learn the representation of $\mathfrak{sl}(3, \mathbb{C})$ through the eigenvalues of H_1 and H_2 . The role of eigenvalues is somehow replaced here by the weight.

4.2.1 Definition. A weight is a pair $\alpha = (\alpha_1, \alpha_2)$ of eigenvalues of $\varphi(H_1)$ and $\varphi(H_2)$ whose have same eigenvector. This eigenvector is called then weight vector associated with α . The set of weight vectors associated to α , V_α is subspace, called weight space. The multiplicity of α is the dimension of V_α .

As, we deal in complex case, $\varphi(H_1)$ admits an eigenvalue α_1 corresponding to eigenspace W . $[H_1, H_2] = 0$ implies $\varphi(H_1)$ and $\varphi(H_2)$ commute, then $\varphi(H_2)(W) \subseteq W$. Considering the restriction of $\varphi(H_2)$ on W , there is eigenvalue α_2 of $\varphi(H_2)$ with eigenspace $W' \subseteq W$. So $\alpha = (\alpha_1, \alpha_2)$ is a weight for (φ, V) . This shows that any representation admits at least one weight.

One can notice also, by the fact the restrictions of φ on $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are isomorphic each to representation of $\mathfrak{sl}(2, \mathbb{C})$, any weight of (φ, V) must be element of $\mathbb{Z}^2$.

Now, we need one more definition to see the action $\varphi(X_i)$ and $\varphi(Y_i)$ on each weight space of (φ, V) .

4.2.2 Definition. A root is an element $r = (r_1, r_2)$ of $\mathbb{C}^2 \setminus \{(0, 0)\}$ such that there is $Z_a \in \mathfrak{sl}(2, \mathbb{C})$ with $[H_i, Z_r] = r_i Z_r$. Such Z_a is called root vector associated with a .

By computation, one can check is, the only roots of $\mathfrak{sl}(3, \mathbb{C})$ are

$$a := (2, -1) \leftrightarrow X_1, \quad (-2, 1) \leftrightarrow Y_1, \quad b := (-1, 2) \leftrightarrow X_2, \quad (1, -2) \leftrightarrow Y_2, \quad (1, 1) \leftrightarrow X_3, \quad (-1, -1) \leftrightarrow Y_3.$$

4.2.3 Lemma. If r is a root associated with root vector Z_r , then, $\varphi(Z_r)$ maps W_α in $W_{\alpha+r}$.

Proof. Let $w \in W_\alpha$. $\varphi(H_i)(\varphi(Z_r)(w)) = r_i \varphi(Z_r)(w) + \varphi(Z_r)(\varphi(H_i)(w)) = (r_i + \alpha_i) \varphi(Z_r)(w)$. $\square$

By abuse of language, let's say the spectrum of H , $\text{spec}(H)$ is just the set of all weights of (φ, V) . Want to have, something analogous to (4.1.1), one can be claimed is

$$V = \bigoplus_{\alpha \in \text{spec}(H)} V_{\alpha}. \quad (4.2.1)$$

Indeed, the direct sum is not $\{0\}$ as the representation should admits at least one weight. Clearly, it is invariant under the action of H_1 and H_2 . To see that is invariant for the remaining basis, it follows from the fact they are root vectors and Lemma 4.2.3.

Now let's introduce a partial order on $\text{spec}(H)$. For two weights α and β , we say $\alpha \leq \beta$ if there are some non-negative p, q such that $\beta - \alpha = pa + qb$. A lowest weight is a meet¹ of $\text{spec}(H)$ that is also element of $\text{spec}(H)$. The order theory says that the meet ε always exists, in our case.

Let's take a moment to think what we can get quickly from Lemma 4.2.3. First, as X_3 and Y_3 are roots associated with $a + b$ and $-a - b$, if α is weight, each of $(\alpha + ka + kb)_{k \in \mathbb{Z}}$ is either weight also, either zero. The same reasoning with the fact that X_1, X_2 and Y_1, Y_2 are roots, we get also if α is weight, then each $(\alpha + ka + lb)_{k, l \in \mathbb{Z}}$ is either weight also, either zero. Using the irreducibility and finiteness,

$$\text{spec}(H) \subseteq \{\varepsilon + ka + lb : k = 0, 1, \dots, N; l = 0, 1, \dots, M\}. \quad (4.2.2)$$

The goal now, is to show that ε is a weight then the lowest weight and the inclusion (4.2.2) is near to be equality. Let $\bar{\varepsilon} = \varepsilon + ma$ such that m is the minimum non-negative integer such that $\varepsilon + ma \in \text{spec}(H)$. Such m exists because ε is the meet. There is a nonzero vector v in $V_{\bar{\varepsilon}}$. Now, notice by minimality of m , $\varphi(Y_1)(v) = 0$. By definition of ε , $\varphi(Y_2)(v) = 0$. As $[Y_1, Y_2] = -Y_3$, we have also $\varphi(Y_3)(v) = 0$.

4.2.4 Proposition. For $v \in V_{\bar{\varepsilon}} \setminus \{0\}$, $\text{span} \left(\{\varphi(X_{i_1})\varphi(X_{i_2}) \cdots \varphi(X_{i_k})(v) : k \in \mathbb{N}, 1 \leq i_j \leq 3\} \right) = V$.

Proof. Let W the span. Clearly, it is invariant under the action of X_i . To see that it is invariant under the action of Y_i and H_i , let's show it by induction on k . For $k = 0$, nothing to for the H_i . For Y_i , have seen above $\varphi(Y_i)(v) = 0$. The induction follows from the fact (can change the role of Y_i by H_i)

$$\varphi(Y_i)(\varphi(X_{i_1})\varphi(X_{i_2}) \cdots \varphi(X_{i_k})) = [\varphi(Y_i), \varphi(X_{i_1})](\varphi(X_{i_2}) \cdots \varphi(X_{i_k})) + \varphi(X_{i_1})\varphi(Y_i)(\varphi(X_{i_2}) \cdots \varphi(X_{i_k})).$$

Each term of the sum reduces to the case $k - 1$. Thus, W is subrepresentation of (φ, V) . $\square$

From this fact, one can conclude also, $V_{\bar{\varepsilon}}$ has dimension one because the roots associated to root vectors X_i increase at least on coordinate, then $V_{\bar{\varepsilon}}$ can not contain $\varphi(X_{i_1})\varphi(X_{i_2}) \cdots \varphi(X_{i_k})(v)$ unless $k = 0$. Another important consequence of this fact is any element in V has weight $\bar{\varepsilon} + pa + qb$ with p, q positive. That implies directly $\bar{\varepsilon} = \varepsilon = (\varepsilon_1, \varepsilon_2)$. Hence the lowest weight always exists.

Using Lemma 4.1.1, $-\varepsilon_1$ and $-\varepsilon_2$ are biggest positive integers such that $\varphi(X_i)^{\varepsilon_i}(v) \neq 0$.

By the same reasoning we can prove also that the highest weight always exists.

4.2.5 Proposition. If ε is the lowest weight and μ is the highest weight Then $\mu_1 = -\varepsilon_2$ and $\mu_2 = -\varepsilon_1$.

Proof. Assume ε is the lowest weight. First, let's see that it $\bar{\mu} = (-\varepsilon_2, -\varepsilon_1)$ is an upper bound of $\text{spec}(H)$. It suffices to notices notice for any $(p > -\varepsilon_1 - \varepsilon_2$ or $q > -\varepsilon_1 - \varepsilon_2)$ and $p, q \geq 0$, $\varepsilon + pa + qb$ can not be a weight². Now, if α is weight, then there is p, q non-negative integer such that

¹that is the biggest among the lower bound (infimum). One can show is, it exists here (by finiteness and irreducibility).

²Either we reason by a little bit heavy computation, either we try to reason the representation of $\mathfrak{sl}(2, \mathbb{C})$ in clever way.

$\alpha = \varepsilon + pa + qb$. Solving, $\bar{\mu} = \alpha + p'a + q'b$, we get $p' = -\varepsilon_1 - \varepsilon_2 - p$ and $q' = -\varepsilon_1 - \varepsilon_2 - q$ that are non-negative. So $\bar{\mu}$ is upper bound.

For now, the simplest way to see that is the join³, we can expect that there is symmetry on diagram of weight between the lowest weight ε and the highest weight μ . So $-\varepsilon_1 - \varepsilon_2 = \mu_1 + \mu_2$ as ε_i and μ_i are the lengths of the side of corners $-\varepsilon$ and μ . But $\mu \leq \bar{\mu}$, then there are non-negative p, q such that

$$\begin{aligned} (-\varepsilon_2, -\varepsilon_1) &= (\mu_1, \mu_2) + p(2, -1) + q(-1, 2) \\ \implies \mu_1 + \mu_2 &= -\varepsilon_1 - \varepsilon_2 = \mu_1 + 2p - q + \mu_2 - p + 2q \implies p = -q \implies p = q = 0. \end{aligned}$$

Hence $\mu = \bar{\mu}$ □

We can rewrite (4.2.2) now

$$\text{spec}(H) \subseteq \{\alpha : \varepsilon \leq \alpha \leq \mu\} = \{\varepsilon + ka + lb = \mu - k'a - l'b : k, k', l, l' = 0, 1, \dots, \mu_1 + \mu_2\}. \quad (4.2.3)$$

The next task is to show that the diagram of weight is full.

So far, each $k = 0, \dots, -\varepsilon_1$, $\varphi(X_1)^k(v) \neq 0$ and $\varphi(Y_2)^k(v) \neq 0$; for each $l = 0, \dots, -\varepsilon_2$, $\varphi(X_2)^l(v) \neq 0$ and $\varphi(Y_1)^l(v) \neq 0$. Also $\varphi(X_1)^{-\varepsilon_1+1}(v) = \varphi(X_2)^{-\varepsilon_2+1}(v) = \varphi(Y_1)^{-\varepsilon_2+1}(v) = \varphi(Y_2)^{-\varepsilon_1+1}(v) = 0$

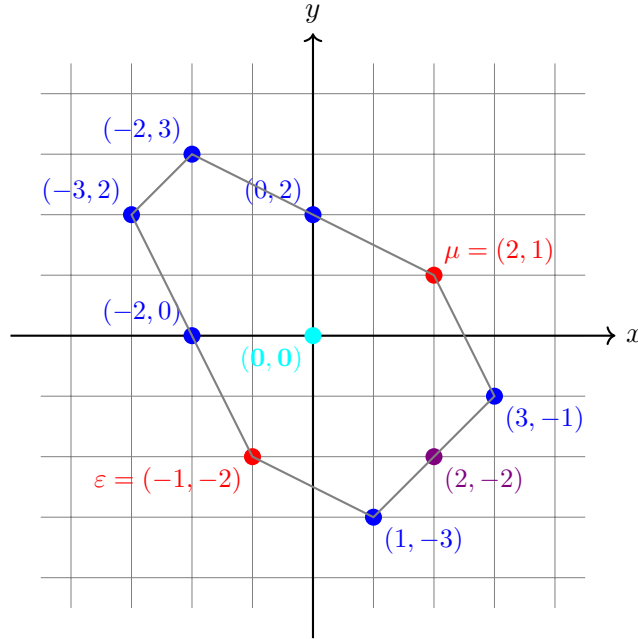


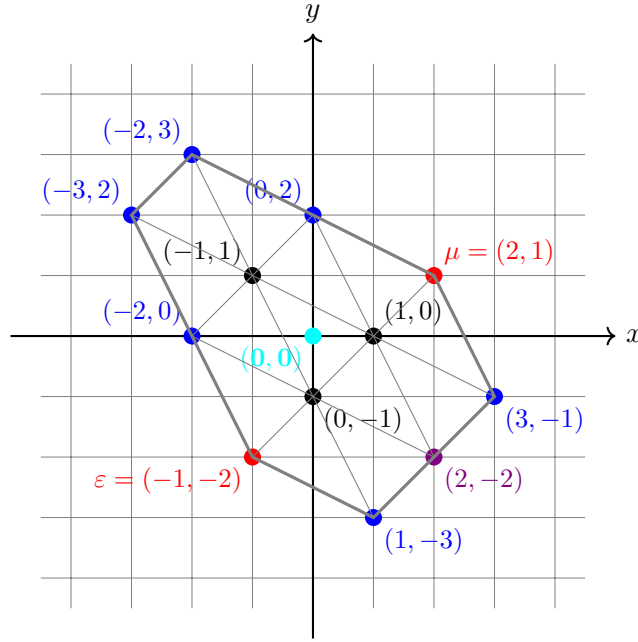
Figure 4.1: Partial weight diagram with lowest weight $\varepsilon = (-1, -2)$.

So far we know that all of point in blue (in the edge) of the hexagon in figure 4.1 are weight, in $\text{spec}(H)$. But the point $(2, -2)$, in purple is not sure yet. One can do it from up.

I claim that $(-1, 1)$ is also weight. Why? because to reach $(-2, 3)$ we have to pass there or the path is of form $X_3X_1X_1(v)$. But saying $(-1, 1) \notin \text{spec}(H)$ implies the only path possible is $X_3X_1X_1(v) \neq 0$, in particular $X_3X_1(v) = 0$. But it is contradiction because $[X_1, X_3] = 0$, then $X_3X_1X_1(v) = X_1(X_3X_1(v)) = 0$.

We can use the same reasoning to complete up to down. So the diagram is full.

³dual of meet (supremum)

Figure 4.2: Full weight diagram with lowest weight $\varepsilon = (-1, -2)$.

Sketch of formal (general case) proof to show that the weight diagram is full.

- (i) Notice that any edges(points) in four segments $[\varepsilon, \varepsilon - \varepsilon_2 b]$, $[\varepsilon, \varepsilon - \varepsilon_1 a]$, $[\mu, \mu + \varepsilon_2 a]$ and $[\mu, \mu + \varepsilon_1 b]$ are all weights.
- (ii) Notice that each of the points in the segment $[\varepsilon - \varepsilon_2 b, \mu + \varepsilon_2 a]$ is weight, using the fact X_3 commutes with X_k and Proposition 4.2.4.
- (iii) Repeat the same reasoning recursively for the next oblique segment until the segment $[\varepsilon - \varepsilon_1 a, \mu + \varepsilon_1 b]$.

□

4.2.6 Proposition. Two irreducible representations having same lowest weight are isomorphic.

Proof. Very similar to the one in Hall (2013) Proposition 6.15.

□

4.2.7 Example. First, the trivial example is $\varepsilon = (0, 0)$, that is $\varphi = 0$.

The next standard one is the of lowest weight $(0, -1)$. That is φ_1 is just the inclusion from $\mathfrak{sl}(3, \mathbb{C})$ to $\mathfrak{gl}(V)$ for $V = \mathbb{C}^3$. We can check that it is irreducible and e_1, e_2, e_3 have weight respectively $(1, 0), (-1, 1), (0, -1)$.

Now, one can see the one with lowest weight $(-1, 0)$ is the dual of the standard above, it can be thought just $\varphi_2(Z) = -Z^T$ for $V = \mathbb{C}^3$. e_1, e_2, e_3 have weight respectively $(-1, 0), (1, -1), (0, 1)$.

By unicity, for the both example, it just check that they are irreducible and they verify the things that we have studied so far.

4.2.8 Proposition. For $\varepsilon_1, \varepsilon_2$ non-positive integers, there is an irreducible representations of $\mathfrak{sl}(3, \mathbb{C})$ that has lowest weight $\varepsilon = (\varepsilon_1, \varepsilon_2)$.

We will discuss the general case in section 4.3.10.

There are many things interesting that we should ask about the properties of the unique irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ of lowest weight ε . The dimension, the multiplicity of each weight.

4.2.9 Theorem. Let V be the irreducible representation of $\mathfrak{sl}(3, \mathbb{C})$ with lowest weight $\varepsilon = (\varepsilon_1, \varepsilon_2)$. Then its dimension is given by

$$\dim V = -\frac{1}{2}(\varepsilon_2 - 1)(\varepsilon_1 - 1)(\varepsilon_1 + \varepsilon_2 - 2).$$

The proof of this formula is seen at [Hall \(2013\)](#). Also the multiplicity of each weight is something already known by Kostant's Multiplicity Formula, seen in chapter 10, section 6, Theorem 10.29, in the same book.

4.3 Representations of complex Semisimple Lie algebra

We have seen from $\mathfrak{sl}(2, \mathbb{C})$, the work is around the diagonalizable matrix H , for $\mathfrak{sl}(3, \mathbb{C})$, we deal with the vectors H_1, H_2 and used the fact that it is diagonalizable to conclude that V is sum of weight spaces. First, Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{g}$ is a maximal abelian subalgebra where any $\text{ad}(H)$ is diagonalizable for each $H \in \mathfrak{h}$. Such subalgebra exists and unique up to isomorphism. The general study of representations of a Lie algebra will revolve around this Cartan subalgebra $\mathfrak{h}$.

Here, by root r , we mean a non trivial element of the dual of $\mathfrak{h}$, $\mathfrak{h}^*$ that has non-trivial root space

$$\mathfrak{g}_r := \{X \in \mathfrak{g} : [H, X] = r(H)X \text{ for all } H \in \mathfrak{h}\}.$$

As $\mathfrak{g}$ is the sum of all common eigenspaces of $\text{ad}(H)$, $X \in \mathfrak{h}$, then $\mathfrak{g} = \mathfrak{h} \oplus \sum_{r \in \Delta} \mathfrak{g}_r$ where Δ is the set of roots. Combining this with the fact $\mathfrak{g}$ is semisimple, the Killing form $B|_{\mathfrak{h}}$ is also non-degenerate, then we can identify $\mathfrak{h}$ with $\mathfrak{h}^*$ by $H_r \in \mathfrak{h} \mapsto B(H_r, -)$ ⁴. We will write $\langle r, s \rangle = \langle \tilde{H}_r, \tilde{H}_s \rangle = r(\tilde{H}_s) = s(\tilde{H}_r)$. By computation one can see easily, is, $[\mathfrak{g}_r, \mathfrak{g}_s] \subseteq \mathfrak{g}_{r+s}$ (here $\mathfrak{g}_0 = \mathfrak{h}$ and $\mathfrak{g}_t = (0)$ if $t \notin \Delta$). Also, one fact is $\mathfrak{g}_r$ is one dimensional subspace.

By definition, the root lattice $\Lambda_{\mathfrak{g}}$ is the lattice in $\mathfrak{h}^*$ generated by the roots.

A r string containing s is the set of all roots (with zero functional) of form $s + i r$ for some $i \in \mathbb{Z}$.

4.3.1 Proposition. Let r be in Δ , and let s be in $\Delta \cup \{0\}$.

- (a) The r string containing s has the form $s + i r$ for $-p \leq i \leq q$ with $p \geq 0$ and $q \geq 0$. There are no gaps. Furthermore

$$p - q = \frac{2\langle s, r \rangle}{\langle r, r \rangle}, \text{ and } \frac{2\langle s, r \rangle}{\langle r, r \rangle} \in \mathbb{Z}.$$

- (b) If $s + i r$ is never 0, define $\mathfrak{sl}_r$ to be the isomorphic copy of $\mathfrak{sl}(2, \mathbb{C})$ spanned by some H'_r , E'_r , and E'_{-r} , and let

$$\mathfrak{g}' = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_{s+i r}.$$

Then the representation of $\mathfrak{sl}_r$ on $\mathfrak{g}'$ by ad is irreducible.

This implies also if $s = c r$, then $c \in \{-1, 0, 1\}$. The goal of this section not provide all detailed proofs but to learn how looks like the representation. The proof is seen in [Knapp and Knapp \(1996\)](#). The key point is there is Lie subalgebra that acts like $\mathfrak{sl}(2, \mathbb{C})$ and we will use this fact to learn its representation. So we choose or renormalize H_r if needed to get good basis. [Hall](#) has more precise description.

⁴Injective then isomorphism since $\mathfrak{h}$ and $\mathfrak{h}^*$ have same dimension(finite)

4.3.2 Proposition (Hall (2013)). For each root r , we can find linearly independent elements X_r in $\mathfrak{g}_r$, X_{-r} in $\mathfrak{g}_{-r}$, and H_r in $\mathfrak{h}$ such that H_r is a multiple of r and such that

$$[H_r, X_r] = 2X_r, \quad [H_r, X_{-r}] = -2X_{-r}, \quad [X_r, X_{-r}] = H_r. \quad (4.3.1)$$

We note that this H_r is not exactly the $\tilde{H}_r$ from the fact B is non-generate, it is normalized one $\frac{2}{\langle r, r \rangle} \tilde{H}_r$.

4.3.3 Definition. $H_0 \in \mathfrak{h}$ is said to be regular if for any $r \in \Delta$, $r(H_0) \neq 0$ ⁵. Let's say $r \leq s$ if $r(H_0) \leq s(H_0)$ and Δ^+ the set of root r with $r(H_0) > 0$. A root r is fundamental if r is not sum of any roots in Δ^+ . We denote $F = \{f_1, \dots, f_m\}$ the set of fundamental roots.

4.3.4 Proposition. (a) If $f_i, f_j \in F$, then $\langle f_i, f_j \rangle \leq 0$ for $i \neq j$.

(b) The elements of F are linearly independent.

(c) If $r \in \Delta^+$, then $r = \sum_{i=1}^m n_i f_i$ with $n_i \geq 0$. In particular, F is a basis of $\mathfrak{h}_{\mathbb{R}}^*$.

Proof. See Ziller (2010). □

The thing is we have all of the roots look like $r^{(1)}, -r^{(1)}, \dots, r^{(l)}, -r^{(l)}$; with root vectors $X_1 := X_{r^{(1)}}, Y_1 := X_{-r^{(1)}}, \dots, X_l := X_{r^{(l)}}, Y_l := X_{-r^{(l)}}$ and $H_{r^{(i)}} := H_i$, $\{X_i, Y_i, H_i\}$ respects (4.3.1). This construction is the main tool that we will use, they exists, the detailed proof is in Hall (2013), Knapp and Knapp (1996). On another point of view the $(r^{(\sigma(i))})_{i=1}^m$ ⁶ are the fundamental roots $(f_i)_{i=1}^m$, by choosing only m (called positive for some condition), such way $r^{(i)}$ is not a sum of other roots. One can show also, the $(H_i)_{i=1}^m$ span $\mathfrak{h}$. Also $\text{rk}(\mathfrak{g}) := \dim(\mathfrak{h}) = m$.

Another important generalisation is the weight of a representation (φ, V) . Again, we assume that it is irreducible and V is finite dimensional. An element $\mu \in \mathfrak{h}^*$ is its weight if it has non-trivial weight space

$$V_\mu := \{v \in V : H \cdot v = \varphi(H)(v) = \mu(H)v \text{ for all } H \in \mathfrak{h}\}.$$

Recall that for any $i = 1, \dots, m$: $\{X_1, Y_1, H_i\}$ is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. Then $\mu(H_i) \in \mathbb{Z}$. As the (H_i) span $\mathfrak{h}$, let's encode the information of $\text{spec}(H)$ in

$$\mathcal{L}(\mathfrak{g}) := \{(\mu(H_1), \dots, \mu(H_m)) : \mu \in \text{spec}(H)\} \subseteq \mathbb{Z}^m.$$

The small clarification is here, $\mathcal{L}(\mathfrak{g})$ is the thing that we call $\text{spec}(H)$ for the $\mathfrak{sl}(3, \mathbb{C})$ case. So an element of $\mathcal{L}(\mathfrak{g})$ is a m -tuple (μ_i) if that there is nonzero vector in V , simultaneously eigenvector of $\varphi(H_i)$ associated to eigenvalue μ_i . This $\mathcal{L}(\mathfrak{g})$ can be viewed exactly $\text{spec}(H)$ because a functional is given by his value on the basis. So, we may confuse them in the next.

The same for the roots Δ , it can be identified with subset of $\mathbb{Z}^m$ by thinking only the image of H_i .

Now, we tried to repeat the step like $\mathfrak{sl}(3, \mathbb{C})$ case with generalization with small adjustment if needed. Making in mind that we treated the problem without talking the Cartan subalgebra $\mathfrak{h}$ but it was involved as we work on H_1 and H_2 that span $\mathfrak{h}$. Now, by the same computation, Lemma 4.2.3 is still true. Then we have again (4.2.1), as $\mathfrak{g}$ is spanned by the $\mathfrak{h}$ and the root vectors.

Usually, one can work on the partial order defined at 4.3.3, but, I want to do the same that I did for $\mathfrak{sl}(3, \mathbb{C})$ case. Then let's define, for weights $\alpha, \beta \in \text{spec}(H)$ $\alpha \preceq \beta$ means $\beta - \alpha = \sum_{i=1}^m n_i f_i$ with

⁵Such element always exists because Δ is finite (but we fix one), and we can choose it so that $\alpha(H_0) \in \mathbb{R}$ for each $\alpha \in \Delta$.

⁶ $\sigma \in S_n$ for some $m \leq l$.

$n_i \geq 0$ and $f_i \in F$. Here, the meet ε of $\text{spec}(H)$ exists and we claim that it is also lowest weight.

Let $k = 1, \dots, m$ and m_k the minimum non-negative integer such that $\bar{\varepsilon} = \varepsilon + m_k f_k$ is weight. Then there is nonzero vector $v \in V_{\bar{\varepsilon}}$. The minimality of m_k says $Y_k \cdot v = 0$. The definition of ε says also $Y_i \cdot v = 0$, for each $i = 1, \dots, m; i \neq k$. As any positive root is sum of simple root then, it the another $Y_j, j > m$ can be expressed as brackets of $X_i, i = 1, \dots, m$. Then for them also $Y_j \cdot v = 0$.

4.3.5 Proposition. For $v \in V_{\bar{\varepsilon}} \setminus \{0\}$, $\text{span} \left(\{ \varphi(X_{i_1}) \varphi(X_{i_2}) \cdots \varphi(X_{i_k})(v) : k \in \mathbb{N}, 1 \leq i_j \leq l \} \right) = V$.

The proof is very similar to 4.2.4. As $\mathfrak{sl}(3, \mathbb{C})$ case, this gives us $V_{\bar{\varepsilon}}$ is one dimensional and $m_k = 0, \bar{\varepsilon} = \varepsilon$.

Since we took an arbitrary k , we can say ε is a weight, then a lowest weight. Using the identification with $\mathfrak{sl}(2, \mathbb{C})$ and Lemma 4.1.1, viewing $\varepsilon \in \mathcal{L}(\mathfrak{g})$, $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ where the components are biggest positive integers such that $\varphi(X_i)^{\varepsilon_i}(v) \neq 0$.

Let's now talk about an important group that acts on set of weight.

4.3.6 Definition. For a root $r \in \Delta^+$, $s_r : \mathfrak{h} \longrightarrow \mathfrak{h}, H \mapsto H - \frac{2\langle H, r \rangle}{\langle r, r \rangle} \tilde{H}_r = H - \langle H, r \rangle H_r$ is reflection in the hyperplane $\{M \in \mathfrak{h}_{\mathbb{R}} \mid r(M) = 0\}$. The Weyl group $\mathbf{W}(\mathfrak{g})$ is the group generated by all of the s_r , $r \in \Delta^+$. The Weyl chamber is $\mathbf{W}_c := \{ \lambda \in \mathfrak{h}_{\mathbb{R}}^* \mid \lambda(H_i) \geq 0 \text{ for all } i = 1, \dots, m \}$.

4.3.7 Remark (Observation). The following facts are tools that can help us better understand the representation of $\mathfrak{g}$. We skip the proofs, which can be found in [Fulton and Harris \(2013\)](#) or [Hall \(2013\)](#).

- (i) s_r induces a reflection on $\mathfrak{h}^*$ and acts the roots $s_r(s) = s - s(H_r) r$.
- (ii) The set of weights of any representation of $\mathfrak{g}$ is invariant under the Weyl group.
- (iii) s_r preserves $\mathfrak{h}_{\mathbb{R}}$.
- (iv) Every element of $\mathbf{W}(\mathfrak{g})$ is induced by an automorphism of the Lie algebra $\mathfrak{g}$ carrying $\mathfrak{h}$ to itself.
- (v) Every vertex of the convex hull of the weights of V must be conjugate to μ under the Weyl group, if μ is the highest weight.
- (vi) $\mathbf{W}(\mathfrak{g})$ acts simply transitively on $\mathbf{W}_c$.

From point (v) Proposition 4.3.1, part (a), $\mathcal{L}(\mathfrak{g})$ is the $\alpha = \mu + r$ for some $r \in \Delta^-$ such that α lie in the convex all of weights.

4.3.8 Theorem (Fulton and Harris (2013)). For any μ in the intersection of the Weyl chamber $\mathbf{W}_c$ associated to the ordering of the roots with the weight lattice Λ_W ⁷, there exists a unique irreducible, finite-dimensional representation Γ_{μ} of $\mathfrak{g}$ with highest weight μ ; this gives a bijection between $\mathscr{W} \cap \Lambda_W$ and the set of irreducible representations of $\mathfrak{g}$. The weights of Γ_{μ} will consist of those elements of the weight lattice congruent to μ modulo the root lattice $\Lambda_{\mathfrak{g}}$ and lying in the convex hull of the set of points in $\mathfrak{h}^*$ conjugate to α under the Weyl group.

The proof of the existence, there is an abstract construction but we will prove only the uniqueness. To do it let's start by a fact.

4.3.9 Lemma. Let (φ, V) be representation of $\mathfrak{g}$, not necessary irreducible. If v is a highest weight vector and $\mathfrak{g} = \mathfrak{g}^+ \oplus \mathfrak{h} \oplus \mathfrak{g}^-$ is the decomposition with respect to the positive and negative root spaces, then the subrepresentation generated by v , $U(\mathfrak{g}) \cdot v$, is equal to $U(\mathfrak{g}^-) \cdot v$, which is the smallest subspace containing v and invariant under the $\mathfrak{g}^-$ -action.

⁷The set of all functionals f in $\mathfrak{h}^*$ where $f(H_i) \in \mathbb{Z}$ for each i .

Proof. Let $w = M_1 M_2 \cdots M_k \cdot v$ and let's prove by induction k that it is in $U(\mathfrak{g}^-) \cdot v$. For $k = 1$, nothing to prove. Let's use the induction for $k \geq 2$. If $M_1 \in \mathfrak{g}^-$, nothing to prove. If $M_1 = H \in \mathfrak{h}$. Note that $[H, A_1 A_2 \cdots A_\ell] = (\theta_1(H) + \cdots + \theta_\ell(H)) A_1 A_2 \cdots A_\ell$ if $[H, A_j] = \theta_j(H) A_j$. It follows that $M_1 M_2 \cdots M_k \cdot v = M_2 \cdots M_k H \cdot v + (\theta_2(H) + \cdots + \theta_\ell(H)) M_2 \cdots M_k \cdot v \in U(\mathfrak{g}^-)$ by induction. Doing the same thing, but , more case and a little computation we can get also the case where $M_1 \in \mathfrak{g}^+$. $\square$

Proof of unicity in Theorem 4.3.8. Let V and W be two irreducible representations of same highest weights μ with weight vectors respective v and w . Clearly (v, w) is highest weight of $V \oplus W$. By Lemma 4.3.9, the subrepresentation of $V \oplus W$ generated by (v, w) , $U = U(\mathfrak{g}) \cdot (v, w)$ is $U(\mathfrak{g}^-) \cdot (v, w)$. The top weight space is dimension one. Now , if $(0) \neq U'$ subrepresentation of U , there is $0 \neq u \in U'$. We can raise it using some elements of $\mathfrak{g}^+$ until we get nonzero vector u' such that for any $X \in \mathfrak{g}^+$, $X \cdot u' = 0$. Then u' is in highest weight space, so, scaling of (v, w) . Hence $U \subset U'$ and U is irreducible. By the Schur's lemma⁸ to conclude that the restriction of the projections on U are isomorphism. $\square$

The fundamental weights $\omega_1, \dots, \omega_m \in \mathfrak{h}^*$ are defined by $\omega_i(H_j) = \delta_{ij}$ for $1 \leq i, j \leq m$. They are all in $\mathbf{W}_c$ and any weights in $\mathbf{W}_c$ can be written uniquely as $\sum_i n_i \omega_i$, $n_i \in \mathbb{N}$. But now, let's try to give an example to see what we did.

4.3.10 Representation of $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{C})$

It is easy to see that the Cartan subalgebra is $\mathfrak{h} = \{H = \text{diag}(h_1, \dots, h_n) : \sum h_i = 0\}$. Indeed it is abelian and $\text{ad}(H)$ maps E_{ij} to $(h_i - h_j)E_{ij}$, where E_{ij} denotes the matrix with a 1 in the i -th row and j -th column and zeros elsewhere, then $\text{ad}(H)$ is diagonalizable. To see that $\mathfrak{h}$ is maximal abelian, if $Y \in \mathfrak{sl}(n, \mathbb{C})$ commutes with $\mathfrak{h}$, by computation, it have to be diagonal matrix in $\mathfrak{h}$.

We define $r^{(ij)} : \mathfrak{h} \longrightarrow \mathbb{C}$, $H \mapsto (h_i - h_j)$. One can check, they are root if $i \neq j$, the associated root spaces $g_{ij} = \langle E_{ij} \rangle$. Also $-r^{(ij)} = r^{(ji)}$. we can think now, $X_{r^{(ij)}} = E_{ij}$. We can compute that $B(X, Y) = 2n \text{Tr}(XY)$. Then the identification of $r^{(ij)}$ in $\mathfrak{h}$, via B is

$$\tilde{H}_{ij} = \frac{1}{2n}(E_{ii} - E_{jj}).$$

One can show also is for each $i \leq n-1$, $r^i := r^{(i, i+1)}$ is fundamental roots and $[E_{i, i+1}, E_{i+1, i}] = E_{ii} - E_{jj} = 2n\tilde{H}_{ij} := H_i$. So $(H_i)_{i=1}^{i=n-1}$ span $\mathfrak{h}$. The $\langle X_i := X_{r^{(i, i+1)}}, Y_i := X_{r^{(i+1, i)}}, H_i \rangle$ is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Let $(\pi, \mathbb{C}^n)$ the standard representation, that is clearly irreducible and of weight

$$\mathcal{L}(\mathfrak{g}) = \{(1, 0, \dots, 0), (-1, 1, 0, \dots, 0), (0, -1, 1, 0, \dots, 0), \dots, (0, \dots, 0, -1, 1), (0, \dots, 0, -1)\},$$

whose the lowest is $(0, \dots, 0, -1)$ and the highest is $(1, 0, \dots, 0)$.

Next, we claim that $\wedge^k \mathbb{C}^n$ is irreducible representation of highest weight $\mu^{(k)} = (0, \dots, 0, 1, 0, \dots, 0)$ (1 only in the k -th component, zeros otherwise). The basis of $\wedge^k \mathbb{C}^n$ is $\{e_{i_1} \wedge \cdots \wedge e_{i_k} : i_1 < \cdots < i_k\}$. By Leibniz rules, each of them has weight $L_{i_1} + \cdots + L_{i_k}$, if $L_j : H \mapsto h_j$. The highest among them is $\mu^{(k)} = L_1 + \cdots + L_k$ ⁹ because $\mu_k - (L_{i_1} + \cdots + L_{i_k}) = \sum_i x_j r^{(j, j+1)}$, for some (x_j) . One can show that is irreducible. Indeed, let W is invariant. Noticing $[E_{a,b}, E_{c,d}] = \delta_{b,c}E_{a,d} - \delta_{a,d}E_{c,b}$, $\mathfrak{b}^+ = \text{span}(E_{ij} : i < j)$ is Lie subalgebra of $\mathfrak{sl}(n, \mathbb{C})$, then we can apply Lie's Theorem 3.2.5 for $\mathfrak{b}^+ \subseteq \mathfrak{gl}(W)$ to have nonzero vector $w \in W$, simultaneously eigenvectors of each $E_{ij} \in \mathfrak{b}^+$. As they are

⁸Morphism of representation between irreducible ones is either zero, either isomorphism.

⁹ $L_i : \mathfrak{h} \longrightarrow \mathfrak{h}, H \mapsto h_i$.

nilpotent, then $E_{ij} \cdot w = 0$. Then w is highest weight vector. We can check that there is no $e_{i_1} \wedge \cdots \wedge e_{i_k}$ has weight $\mu^{(k)}$ another that $v_k = e_1 \wedge e_2 \wedge \cdots \wedge e_k$ (by the fact the only linear relation among the L_i restricted to $\mathfrak{h}$ is $\sum_{i=1}^n L_i = 0$). It follows $w = \lambda v_k$, $\lambda \in \mathbb{C}$. That mean $v_k \in W$. To conclude $W = \wedge^k \mathbb{C}^n$, let's prove that we can reach any $e_{i_1} \wedge \cdots \wedge e_{i_k}$ using only v_k and the action of $\mathfrak{sl}(n, \mathbb{C})$. It suffices to see $E_{i_k, k} \cdot v_0 = e_1 \wedge \cdots \wedge e_{k-1} \wedge e_{i_k}$, $E_{i_{k-1}, k-1} \cdot (E_{i_k, k} \cdot v_0) = e_1 \wedge \cdots \wedge e_{k-2} \wedge e_{i_{k-1}} \wedge e_{i_k}$. Again and again¹⁰ until $E_{i_1, 1} \cdots E_{i_k, k} \cdot v_0 = e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k}$.

4.3.11 Theorem. For any $\mu \in \mathbb{N}^{n-1}$, there is an unique irreducible representation of highest weight μ ,

$$\Gamma_\mu \subseteq W(\mu) = \text{Sym}^{\mu_1}(V) \otimes \text{Sym}^{\mu_2}(\wedge^2 V) \otimes \cdots \otimes \text{Sym}^{\mu_{n-1}}(\wedge^{n-1} V).$$

Proof. First, highest weights add under tensor products and scale under symmetric powers. Then $W(\mu)$ has highest weight μ . Let v be a weight vector of μ . We claim that if $U(\mathfrak{g})$ universal enveloping algebra of $\mathfrak{g}$, $U(\mathfrak{g}) \cdot v$ ¹¹ does the job. To see it we just need to prove the Poincaré–Birkhoff–Witt (PBW) theorem¹², for $\mathfrak{sl}(n, \mathbb{C})$ case, any element $X \in U(\mathfrak{sl}(n, \mathbb{C}))$ can be written as a finite sum, $X = \sum_j x_-^{(j)} x_0^{(j)} x_+^{(j)}$, with $x_-^{(j)} \in U(\mathfrak{b}^-)$, $x_0^{(j)} \in U(\mathfrak{h})$, $x_+^{(j)} \in U(\mathfrak{b}^+)$. If v is a highest-weight vector, then each $x_+^{(j)}$ kills v , and each $x_0^{(j)}$ acts by a scalar; then $X \cdot v = \sum_j c_j x_-^{(j)} \cdot v$, $c_j \in \mathbb{C}$. So $U(\mathfrak{sl}(n, \mathbb{C})) \cdot v = U(\mathfrak{b}^-) \cdot v$ that is irreducible. Indeed, if T is invariant subspace of $U(\mathfrak{b}^-) \cdot v$ containing nonzero vector u , we can act successively by element of $\mathfrak{b}^+$ to reach some $\bar{u} \neq 0$ such way for all $X \in \mathfrak{b}^+$, $X(\bar{u}) = 0$. Let λ be the weight of $\bar{u}$ and $\mu - \lambda = \sum_{i=1}^{n-1} c_i r^i$, for some $c_i \in \mathbb{N}$; viewing them in each i -th copy of $\mathfrak{sl}(2, \mathbb{C})$, $\langle E_{i,i+1}, H_i, E_{i+1,i} \rangle$ inside $\mathfrak{sl}(n, \mathbb{C})$, each c_i has to be zero. Then $\bar{u}$ and v has same weight then must be proportional, as the highest space¹³ of $U(\mathfrak{b}^-) \cdot v$ is of dimension one. So T contains $U(\mathfrak{b}^-) \cdot v$. $\square$

Theorem 4.1.4 requires μ in the Weyl chamber $\mathbf{W}_c$ to be considerable as highest weight for some irreducible representation. For $\mathfrak{sl}(n, \mathbb{C})$, it is clearly $\mathbf{W}_c = \{\lambda \in \mathfrak{h}_{\mathbb{R}}^* | \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n\}$. This is an identification with $\mathbb{N}^{n-1}$, they are the same, just different point of view. Also, $r^{(ij)} = L_i - L_j$ and after computation $s_{r^{(ij)}}$ is the transposition (L_i, L_j) . So, the Weyl group of $\mathfrak{sl}(n, \mathbb{C})$ is S_n .

4.3.12 Representation of $\mathfrak{g} = \mathfrak{sp}(n, \mathbb{C})$

Recall that $\mathfrak{sp}(n, \mathbb{C}) = \left\{ \begin{pmatrix} B & S_1 \\ S_2 & -B^T \end{pmatrix} \mid B \in \mathfrak{gl}(n, \mathbb{C}), S_i^T = S_i \right\}$. Like we did with $\mathfrak{sl}(n, \mathbb{C})$, one can

show $\mathfrak{h} = \{H = \text{diag}(h_1, \dots, h_n, -h_1, \dots, -h_n) \mid h_i \in \mathbb{C}\}$ is Cartan subalgebra. Let

$M_{k,l} = E_{n+k,l} + E_{n+l,k}$, $N_{k,l} = E_{k,n+l} + E_{l,n+k}$, for $k \leq l \leq n$ and $X_{k,l} = E_{k,l} - E_{n+l,n+k}$, for $k \neq l$, $k, l \leq n$. Then $[H, M_{k,l}] = -(h_k + h_l)M_{k,l}$, $[H, N_{k,l}] = (h_k + h_l)N_{k,l}$, $[H, X_{k,l}] = (h_k - h_l)X_{k,l}$.

These relation tell us the roots are $\pm s^{(ij)} : \mathfrak{h} \rightarrow \mathbb{C}$, $H \mapsto \pm(h_i + h_j)$ for $i \leq j \leq n$ and $r^{(ij)} : \mathfrak{h} \rightarrow \mathbb{C}$, $H \mapsto (h_i - h_j)$ for $i \neq j \leq n$. The root space $\mathfrak{g}'_{ij}$ associated to root $s^{(ij)}$ is span by M_{ij} , $\mathfrak{g}''_{ij}$ associated to root $-s^{(ij)}$ is span by N_{ij} and $\mathfrak{g}_{ij}$ associated to root $r^{(ij)}$ is span by X_{ij} . We can compute that $B(X, Y) = (2n+2) \text{Tr}(XY)$. Then the identification of $s^{(ij)}$ and $r^{(ij)}$ in $\mathfrak{h}$, via B are

$$\tilde{K}_{ij} = \frac{1}{2n+2}(K_i + K_j) \text{ and } \tilde{H}_{ij} = \frac{1}{2n+2}(K_i - K_j), \text{ where } K_i = X_{ii}.$$

Clearly $F = \{r_1 := r^{(1,2)}, \dots, r_{n-1} := r^{(n-1,n)}, r_n := s^{(n,n)}\}$ is linearly independent and we can get every root using them, then, by some choice of regular element, F can be taken as fundamental roots.

¹⁰We used that $E_{ab}e_b = e_a$, $E_{ab}e_j = 0$ for $j \neq b$, and the Leibniz rule on wedges.

¹¹smallest subspace invariant under the action of $\mathfrak{g}$ and containing v .

¹²The idea is the similar to Lemma 4.3.9.

¹³whose has highest weight

Now, let $X_i := X_{i,i+1}$, $Y_i := X_{i+1,i}$ for $i \leq n-1$ and $X_n := N_{n,n}$, $Y_n := M_{n,n}$. We can check by computation that if $H_i = [X_i, Y_i]$ ¹⁴, the $\langle X_i, Y_i, H_i \rangle$ is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$ and $(H_i)_{i=1}^{i=n-1}$ span $\mathfrak{h}$.

Consider the representation of $\mathfrak{sp}(n, \mathbb{C})$ on $W = \mathbb{C}^{2n}$, π by $X \cdot v = Xv$. It is irreducible with weight

$$\mathcal{L}(\mathfrak{g}) = \{\pm w^{(k)} \mid 1 \leq k \leq n\}, \quad w^{(1)} = (1, 0, \dots, 0), \quad w^{(k)} = (0, \dots, 0, -1, 1, 0, \dots, 0), \quad w^{(n)} = (0, \dots, 0, -1, 1).$$

The highest weight is now $w^{(1)}$.

Now, let think about $\pi_k = \wedge^k W$. It is not irreducible for $2 \leq k \leq 2n-2$. Indeed, choosing a symplectic invariant form $\omega \in \wedge^2 W$, one can define $L : \wedge^{k-2} W \rightarrow \wedge^k W$ by $\alpha \mapsto \omega \wedge \alpha$. We claim that the image of L is invariant under the action and it is not zero map, then we have a subrepresentation that is neither zero, neither $\wedge^k W$. By duality, we should expect that it is irreducible for the case $n = 2n-1$ and $2n$; for $\mathfrak{sp}(n, \mathbb{C})$, $\wedge^{2n-1} W \cong \wedge^1 W = W$.

This representation has weights $\mathcal{L}(\mathfrak{g}) = \{\pm w^{(i_1)} \pm \dots \pm w^{(i_l)} \mid i_1 < \dots < i_l \leq k, \quad k-l \text{ even}\}$.

Define $K : \wedge^k W \rightarrow \wedge^{k-2} W$ by $K(v_1 \wedge \dots \wedge v_k) := \sum_{1 \leq i < j \leq k} \omega(v_i, v_j) v_1 \wedge \dots \widehat{v_i} \dots \widehat{v_j} \dots \wedge v_k$; ($\omega(Mu, v) + \omega(u, Mv) = 0$). We claim that $\Gamma_k := \text{Ker } K$ is $\mathfrak{g}$ -invariant. Let $x \in \text{ker } K$, $M \in \mathfrak{sp}(n, \mathbb{C})$.

$$\begin{aligned} K(\pi_k(M)(v_1 \wedge \dots \wedge v_k)) &= \sum_{p=1}^k K(v_1 \wedge \dots \wedge Mv_p \wedge \dots \wedge v_k) = \sum_{i < j} \omega(v_i, v_j) \pi_{k-2}(M)(v_1 \wedge \dots \widehat{v_i} \dots \widehat{v_j} \dots \wedge v_k) \\ &\quad + \sum_{i < j} (\omega(Mv_i, v_j) + \omega(v_i, Mv_j)) v_1 \wedge \dots \widehat{v_i} \dots \widehat{v_j} \dots \wedge v_k = \pi_{k-2}(M) K(v_1 \wedge \dots \wedge v_k). \end{aligned}$$

Therefore, for $x \in \text{ker } K$, $K(\pi_k(M)x) = \pi_{k-2}(M) K(x) = 0 \Rightarrow \pi_k(M)x \in \text{ker } K$.

Moreover, it is irreducible and one can show is, it has highest weight $\mu_k(H) = h_1 + \dots + h_k$. for $k \leq n-1$.

So we have now the irreducible representation of highest weight, for the fundamental weights.

4.3.13 Corollary. For any $\mu \in \mathbb{N}^{n-1}$, there is an unique irreducible representation of highest weight μ ,

$$\Gamma_\mu \subseteq W(\mu) = \text{Sym}^{\mu_1}(W) \otimes \text{Sym}^{\mu_2}(\Gamma_2) \otimes \dots \otimes \text{Sym}^{\mu_n}(\Gamma_n).$$

The proof is similar, what we did with 4.3.11.

¹⁴It is $(2n+2)\tilde{H}_{ii}$ for $i \leq n-1$ and $(2n+2)\tilde{K}_{ii}$ for $i = n$.

5. Final Remarks

The Lie algebra of a Lie group encodes a great deal of information about the group itself. In many cases, the classification of Lie algebras leads directly to the classification of Lie groups up to local isomorphism. Although this project did not cover the full classification theory, it is worth noting that complex semisimple Lie algebras are already completely classified via Dynkin diagrams, a remarkably elegant combinatorial tool. Due to time constraints, we have not presented the theory of Dynkin diagrams in detail.

The study of Dynkin diagrams, together with the theory of the Weyl group, provides the complete classification of complex semisimple Lie algebras. These results also shed light on the classification of real semisimple Lie algebras and, consequently, on semisimple Lie groups. In particular, the passage from complex to real forms involves understanding the structure of non-compact real forms, a subject rich with both algebraic and geometric insights.

So this thesis is just a few starting of the study but we expect to be able to do similar thing for reals onces and another interesting class of Lie algebras and Lie groups.

While this thesis focused on the interplay between Lie groups, their Lie algebras, and their representation theory, such things remains to explore. Also, studding the representation theory with plethysm can be more combinatorial, that is also very interesting. The tools developed here geometric analysis via left-invariant metrics, the study of exponential maps, and the use of representation theory form a solid foundation for deeper investigations into the classification and structure theory of Lie groups. Knowing the curvatures may helps us to understand the Lie groups and vis-versa.

Further directions include studying the role of Cartan connections in the classification of homogeneous spaces and symmetric spaces, exploring the relation between curvature and representation theory, and applying these ideas to the theory of real semisimple Lie groups. Also, having Cartan connection, we can study the geodesics in therm of the algebraic concepts: exponential map.

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